ALL MODELS ARE WRONG, BUT SOME ARE USEFUL.

GEORGE E.P. BOX

THERE IS NO SHORTCUT TO TRUTH, NO WAY TO GAIN KNOWLEDGE OF THE UNIVERSE EXCEPT THROUGH THE GATEWAY OF THE SCIENTIFIC METHOD.

ANTOINE DE SAINT-EXUPÉRY

ERIN E GABRIEL AND PAOLO FRUMENTO

EDITED BY ERIN E GABRIEL, PAOLO FRUMENTO, MICHAEL SACHS, IULIANA CIOCANEA-TEODORESCU, PAUL W DICKMAN, THERESE ANDERSSON, MARK CLEMENTS, ARVID SJÖLANDER, NIKOLAOS SKOURLIS

BIOSTATISTICS I

EPIDEMIOLOGY PHD PROGRAM KAROLINSKA INSTITUTET

Copyright © 2020 Erin E Gabriel and Paolo Frumento
Edited by Erin E Gabriel, Paolo Frumento, Michael Sachs, Iuliana Ciocanea-Teodorescu, Paul W Dickman, Therese Andersson, Mark Clements, Arvid Sjölander, Nikolaos Skourlis

PUBLISHED BY EPIDEMIOLOGY PHD PROGRAM KAROLINSKA INSTITUTET

TUFTE-LATEX.GOOGLECODE.COM

Licensed under the Apache License, Version 2.0 (the "License"); you may not use this file except in compliance with the License. You may obtain a copy of the License at http://www.apache.org/licenses/LICENSE-2.0. Unless required by applicable law or agreed to in writing, software distributed under the License is distributed on an "AS IS" BASIS, WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied. See the License for the specific language governing permissions and limitations under the License.

First printing, April 2020

Contents

Scientific Method 13

Describing Data with Summary Statistics and Graphics 17

Probability 33

Random Variables and Distributions 35

Single Sample Hypothesis Testing and Inference for the Expectation 45

Two sample estimation and inference 61

Causal inference, confounding, mediators and effect modification 75

Linear Regression is Everything 79

Index 103

List of Figures

1 A dotplot shows the individual numeric values as the length along the x-axis. Unless the entries have some inherent ordering, then it is preferable to sort according to the numeric variable. 27

2 Histograms or Density plots for single numeric variables 28

3 A marginal boxplot of a numeric variable will give you summary statistics of the variable. For small to medium datasets, it is good to overlay the raw data points on the boxplot. 28

4 Pie charts show frequencies or percentages of categorical data as the areas of a circle. Dot charts or bar charts are preferred methods of visualizing this sort of information. The axes of the bar charts start at 0, whereas the dot charts do not necessarily. 29

5 Boxplots naturally extend to displaying the association between a numeric variable and a categorical variable. 30

6 Scatterplots show the association between two numeric variables. Smoothing or regression functions can be overlayed to show summary statistics. Individual points or outliers can be highlighted with direct labels. 31

7 Three or more variables can be displayed on a plot in many ways. A third categorical variable can be encoding as color or shape, or used to create subplots in separate panels. Get creative, but remember the general principles. Consider the main message of the visualization, and the cognitive burden of matching colors or shapes to information. Three dimensional figures should be avoided. 31

8 The pmf of a Bernoulli 0.3, the y-axis is the probability of the x value on the x-axis. This is directly given by the Bernoulli pmf 35

9 The pmf of a discrete uniform 6. the y-axis is the probability of the x value on the x-axis. This is directly given by the Uniform pmf, which is the same value for all values, here it is /6 and is indicated by the red line. 36

10 The pmf of a Binomial 5, p=0.4. The y-axis is the probability of the x value on the x-axis. This is directly given by the pmf. 36

11 The pmf of a Poisson 3.2. The y-axis is the probability of the x value on the x-axis. This is directly given by the pmf. The possible values actually continue up to ∞, but the probabilities become very very small after around 10. 37
12 The pdf of a continuous uniform from 1 to 4. 37
13 The pdf of an exponential 2. 38
14 The pdf of a normal with mean 0, and standard deviation 1, in black, 2 in red and 3 in green. 38
15 As $\nu \to \infty$, the T goes to a standard normal. 39
16 Calculating the probability of a set of responses requires calculating the area 40
17 Tail area probabilities for the standard normal distribution. 40
18 Computing probabilities as areas under the normal distribution. 41

19 An Unbiased estimator of the expectation on the bottom and a biased estimator on the top 47
20 A consistent estimator of the expectation with a sample size of 1000 on the top and a sample size of 10 on the bottom 47

21 The permutation distribution of the null 72

22 There is a causal effect of x on y, but there is confounding of the relationship between x and y by c. 76
23 There is a causal effect of x on y, there is no confounding of the relationship between x and y. 76
24 There is a causal effect of x on y, there is no confounding of the relationship between x and y, but there is a mediator m between x and y 76
25 There is a causal effect of x on y, there is no confounding of the relationship between x and y, but there is a different variable q that also impacts y, this may be effect modification. 76

26 Linear regression of SBP on Age, assumes that for a one unit change in age, there is always the same expected difference in SBP. 87
27 For an interaction model the slope over age can be different between males and females. Here they don't go in opposite directions, but the females slope is much lower. 95
28 For an adjusted model gives the same slope over age regardless of sex, but allows for different intercepts by gender. 96
29 The estimated expected value of y will be linear on the log scale, but will not be linear on the original x scale. x has a range 0 to 1000, while the range of $\log_e(\mathrm{x})$ is much smaller. 98
31 Residual versus Fitted value plots to check assumptions 100

List of Tables

1 Examples from each distribution 43
2 Parameters and Notation for the Distributions we have discussed 43

3 Testing Errors type I and type II errors 57

4 Example two-by-two table 66
5 Summary of Two Sample Tests 73

6 The example coefficient matrix from a linear regression with on binary indicator. 84
7 The example coefficient matrix from a linear regression with a single categorical predictor. 85
8 The example coefficient matrix from a linear regression for a single continuous predictor. 86

Introduction

These notes are to be used as a reference during the course.

The Scientific Method and Statistics for Science

Statistics is used in science to aid in the generation of evidence.

How do we generate evidence? We generate evidence using the scientific method. There is no other way of generating evidence.

The Scientific Method

1. Gather information and resources or use existing resources to form a hypothesis [1]
2. Gather data, e.g., by performing an experiment or compiling data from registers or online databases, in which to test a testable implication of your hypothesis. [2]
3. Test the hypothesis in these data. [3]
4. Draw conclusions based on these results and the context of your hypothesis, data collection and testing procedure. [4]
5. Make results public so that others do not have to run the same experiment and so that you may gain feedback and insight as to any bias, errors or other problems that may have lead to your conclusions of than the hypothesis being true or false.
6. Repeat as needed until the scientific community is convinced.

[1] Alternatively you can just take a guess

[2] These data MUST be independent of any data used to form said hypothesis

[3] Statistics might not be the only way. In the testing of a physical law, statistics probably is not needed. Also note that estimation is a form of testing a hypothesis.

[4] This will potentially generating a new hypothesis, rather than providing evidence for or against the original. That is 100% ok, but then you are back at step 1....NOT at step 4

Whenever you do not yet have a specific hypothesis you are at step 1. For example, you test to see what biomarker has the largest mean difference between HIV positive and negative subjects in a cohort study, regardless of how significant your p-value, you are at step 1. You cannot provide evidence to support a hypothesis when you have yet to state one. In addition, even when a hypothesis has been stated, you collect data to test it, and then you do exploratory analysis, for any findings generated in this way, you are at step 1. Exploratory analysis for hypothesis generation and updating is

very important, selling the findings from this analysis as conclusions about the generated hypothesis is why 90% of medical research papers are not reproducible. Finding a significant p-value is not evidence of anything without context. Let's give it some context.

STATISTICAL INFERENCE, for example formal hypothesis testing and p-values, can be misleading in the wrong context. You must think about what you want to test, the hypothesis, while you are collecting the data. Study design and proper data collection are of course statistical methods as well, but there is very little math. There is however what is often referred to as statistical thinking. Considering the sources of bias in sampling and experiments in medical and epidemiological studies under the umbrella of biostatistics, although it is often not discussed in introductory biostats courses, due to time constraints, it is a very critical part of how statistics are properly used in science. Let us consider the steps of the scientific method with an example to see why.

YOU HAVE EXPLORED a random sample from a register and after running some feature selection (machine learning stuff) or just a bunch of logistic regression you find an association, even after adjustment for a lot of other stuff, between low vitamin D levels and increase odds of lung cancer. From this you cannot conclude that low vitamin D levels causes lung cancer, nor should you, as you were fishing around, even concluding that the lung cancer is associated with smoking outside your data. Instead, this generates a hypothesis, but what hypothesis?

HYPOTHESIS ARE ALMOST ALSO causal and must be specific. Asking what is associated with lung cancer is not a hypothesis, it might be a question that helps generate hypotheses, but it is not a hypothesis. 'low vitamin D is associated with an increased risk lung cancer.', is a specific hypothesis, but is this really something you actually want to know. Does this further science? Probably not, but maybe. Ask yourself

- What is my intent with this information?
  1. my intent is to simply tell people about the association, have them follow it up and not imply that low vitamin D causes lung cancer or that stopping smoking would have any effect on lung cancer
  2. my intent is to imply that low vitamin D increases the risk, causal effects, impacts, or changes the risk of lung cancer

If your intent is 1 then this hypothesis is fine as stated. You may wish to refine it to include a group or specific population of interest.

If any of the versions of number 2 are you intent, then this is the wrong hypothesis. Low vitamin D causes lung cancer is the correct hypothesis. Knowing the true hypothesis, helps you know how to collect the data. If this is your hypothesis, then you should consider a well designed and well run clinical trial where you randomize people to low vitamin D or not, and blind to low vitamin D in someway. Although this is possible with Vitamin D, randomized clinical trials are not always possible, and a poorly run randomized clinical trial is no better than, and can actually be much worse than, a well run observational study.

As we have just seen, in step 2 when you are gathering data you must already have the hypothesis in order to gather data in a way that will allow to you properly test the hypothesis and unbiasedly estimate the effect of interest (or association) of interest. There is a lot of deeper causal theory here that is beyond the scope of this course, but we are not going to ignore the implications of this theory. You must sample, gather information, run an experiment in a manner that, at least under some set of assumptions, identifies a testable implication of your hypothesis. Going back to the example, a testable implication of your hypothesis is that on average smoking increases the risk of lung cancer. The ideal study design for testing this would be to randomize all people born over a 1 year period at birth to either low vitamin D or not normal or maybe high vitamin D levels for life, blinding by giving some type of placebo, and then following subjects in each group until everyone gets lung cancer or dies in a different way.

Clearly this experiment is not possible, because you cannot randomize people to something for life and there is no way to randomize everyone. Still even in this ideal setting assumptions would be needed and need to be stated, particularly that this year you picked is not different in some way that makes smoking have an effect when in fact the data generating mechanism for all other years of birth on earth are different. In the real world, other methods of collecting data that are possible must be used. Experimental design and observational study design are fields in their own right and we will not get into them here. However, it should be noted that regardless of the ability to randomize or directly control the exposure of interest, here smoking, and therefor weather or not you are in the experimental or observational study design setting, does not change the fact that you must have a hypothesis and some way to link the hypothesis, or a testable implication of it, to the data that are to be collect PRIOR to its collection. In addition, although it should be clear from the fact the one first sets

the hypothesis and then collects the data, that the ability to collect a particular type of data should not drive the hypothesis. If you are interested in telling people to stop smoking if it increases the risk of lung cancer, your hypothesis is causal, regardless of your ability to collect data that can be used to provide evidence for or against a testable implication of that hypothesis.

NOW WE HAVE DATA OR KNOW WHAT WE ARE GOING TO COLLECT and a testable implication of our hypothesis. Now there are a bunch of statistical details about the data, is it binary or numeric, is it censored or truncated, and of the sample, is it large or small, that need to be considered in order to select an appropriate test or method of estimation. The details of this course are for this step, deciding what test to use and understanding what you are getting from particular test, so that you can provide results that can be used along with the context of the data and testing to draw conclusions in the step 4. Without knowing the pre-specified and testable implication of that hypothesis along with assumptions made about the underlying nature of the data generating mechanism and the collected the data to test said testable implication, we cannot move to step 4, here. However, just because we are using statistics, does not mean we are in step 3.

IN WHAT FOLLOWS WE WILL ASSUME THAT YOU HAVE STATED AND ARE TESTING AN IMPLICATION OF A HYPOTHESIS OR ARE GENERATING A HYPOTHESES USING STATISTICS. This course is concerned with the actual testing and estimation often referred to as statistical inference that one might use in step step 3 to provide evidence to support a hypothesis. However, these same methods can be used in step 1 to generate hypothesis, there is not difference in the methods or the underlying theory supporting them. As well, we will discuss Frequentest statistics, the most popular but not the only framework for statistics, which is mostly considered with the quantification of uncertainty the arises from sampling, mostly ignoring how the sampling was done or any other source of bias in estimation or testing. For this reason, this is not a course in all statistics for science or how statistics can be used in science in general, this is an introduction to one of the statistical approaches to step 1 or step 3 of the scientific method, nothing more. [5]

[5] But honestly that is a lot!

Describing Data with Summary Statistics and Graphics

Introduction

Types of variables

- Categorical (e.g. low/medium/high).
- Binary/dichotomous (e.g., 0/1, M/F) is a special case of categorical.
- Numerical discrete (e.g., number of people in a room, number of events within a set period of time).
- Numerical continuous (e.g., temperature, weight).

Descriptive statistics

- Measures of central tendency (location, position): mean (proportion = mean of a 0/1 variable), quantiles (median, quartiles, min, max, percentiles, proportions).
- Dispersion (variability): range = max − min, interquartile range = 3rd − 1st quantile, standard deviation (σ), variance (σ^2).
- Measures of association: correlation coefficient, regression coefficient, chi-squared, odds ratios, relative risks.
- Measures of asymmetry: skewness, kurtosis, Gini index .
- Many others.

Plots

- Scatterplot (2 numerical variables)
- Boxplot (1 numeric, 1 binary or categorical variable)
- Histogram and density (numerical variable)
- Many others!

Types of variables

EXAMPLE - CATEGORICAL VARIABLE
$z = \{phd, ma, ba, ma, phd, ma, ba, ba, ba, ma\}$

EXAMPLE - BINARY VARIABLE
$y = \{0, 1, 0, 0, 0, 1\}$

EXAMPLE - DISCRETE NUMERICAL VARIABLE
$x = \{0, 0, 1, 2, 3, 4, 5, 6, 9, 12, 40\}$

EXAMPLE - CONTINUOUS NUMERICAL VARIABLE
$q = \{0.3452, 1.567, 1.5, 2.003, 3.213, 4.9876, 5.24, 6.2123, 9.347, 12.098765, 43.98676\}$

USING THE x VARIABLE AS AN EXAMPLE each subject among the 11 sampled, (n=11), is indexed by i, for i in 1 to 11. So, x_1 is equal to 0 and is the value of x for subject 1. This notation will be useful for defining the summary statistics.

Basic summary statistics for a single variable

Numerical variable

THE SAMPLE MEAN is the most basic central summary statistic. It is often denoted by a bar over the variable name. For some variable a, $\overline{a}$ is the sample mean of a. As we will discuss in detail in later chapters, this is a good estimate of the expected value of the underlying data generating mechanism (DGM) which produced the sample, this is simply the true mean of the DGM. The expectation of the DGM is often denoted as μ, and therefore the sample mean is often denoted as being the estimate of this $\widehat{\mu}$

$$\text{Sample mean} = \overline{a} = \widehat{\mu} = \frac{1}{n}\sum_{i=0}^{i=n} a_i = \frac{1}{n}(a_1 + a_2 + \ldots + a_n)$$

THE SAMPLE VARIANCE is a measure of the dispersion about the sample mean. Just like the mean, which is an estimate of a true DGM expectation, the sample variance is an estimate of the true DGM dispersion about the expectation. The estimator for this, if the expectation is not known is:

$$\text{Sample Variance} = \widehat{\text{Var}}(a) = \frac{1}{n-1}\sum_{i=0}^{i=n} (a_i - \overline{a})^2$$

If μ is known, which it will never be, the estimator is given by: [6]

[6] in very large samples even when μ is unknown these will be basically the same, as for very large n, n-1 and n are very close.

$$\text{Sample Variance} = \widehat{\text{Var}}(a) = \frac{1}{n}\sum_{i=0}^{i=n}(a_i - \mu)^2$$

$$\text{Sample Standard deviation} = \widehat{\text{sd}}(a) = \sqrt{\widehat{\text{Var}}(a)}$$

Other basic summaries are the minimum, the maximum and the range of the observed data. Although these are estimates of the min, max and range of the DGM, they are not generally considered very good estimates in the numeric variable case, particularly for continuous variables, as most possible underlying DGM distributions [7] are unbounded, meaning they have a range that includes either positive or negative infinity, or both.

[7] We will get to distributions in Chapter 3

$$\text{sample min} = \text{the smallest observed value}$$

$$\text{sample max} = \text{the largest observed value}$$

$$\text{sample range} = \text{max} - \text{min}$$

Other Summaries we will use and discuss but that are less easy to calculate include the quartiles, including the 25th, median, 75th. These are special versions of the percentiles, which are simply cut-points in the observed that that but the given percentage of the data below that point.

$$\text{sample median} = \text{50th percentile}$$

50% of the data is less than the median. There are a lot of different estimators for this summary. [8]

[8] We will not go into them, but most standard software packages do something reasonable.

$$\text{sample 1st quartile} = \text{sample 25th percentile}$$

[9]

[9] median of values below median

$$\text{sample 3rd quartile} = \text{sample 75th percentile}$$

[10]

[10] median of values above median

Looking at the min and max in this way, we can say:

$$\text{sample min} = \text{0th percentile}$$

$$\text{sample max} = \text{100th percentile}$$

Categorical and Binary Variables

FOR BINARY DATA, the sample mean is still a useful summary, but it is normally referred to as the proportion of 1s, or the proportion of subjects with the indicated characteristic. Just like the sample mean in the numeric variable case, the proportion is not only a summary, but is an estimate of the probability of a variable, let's call is y, being equal to 1. This is often denoted as $p = P(Y = 1)$

$$\text{Proportion} = \text{Sample mean} = \widehat{p} = \frac{1}{n}\sum_{i=0}^{i=n} y_i$$

THE SAMPLE VARIANCE is also meaningful here, [11]

$$\text{Sample Variance} = \widehat{\text{Var}}(a) = \frac{1}{n-1}\sum_{i=0}^{i=n}(y_i - \widehat{p})^2$$

[11] One can also use $\widehat{p} \times (1 - \widehat{p})$ when considering binary data, these should be equivalent in large enough sample sizes and is always equal to the known μ estimator *try it!*. This is a potentially better estimates of the true DGM variance for binary data even is small samples, we will discuss this later in Chapter 6

FOR CATEGORICAL DATA, counts of subjects within each category or proportions within the categories are the most common summaries. This is often given as the table, using the z data above as an example a frequency table of these data would look like this:

ba	ma	phd
4	4	2

and a proportion table would look like this:

ba	ma	phd
.4	.4	.2

z	ba	ma	phd
phd	0	0	1
ma	0	1	0
ba	1	0	0
ma	0	1	0
phd	0	0	1
ma	0	1	0
ba	1	0	0
ba	1	0	0
ba	1	0	0
ma	0	1	0

CATEGORICAL DATA can be, and generally must be, converted to binary data in order to use it in any of the forms of statistical analysis we will discuss below. Again using z as any example we would create 3 such binary variables, ba, ma, and phd. They would be indicator variables, that take on the value 1 if a subject has such a degree as there highest degree and 0 otherwise. One would use this set indicator variables in place of z. [12]

[12] these are also sometimes call 'dummy' variables

Rates

Events over a set period of time are binary variables, but they have an time attached to them. For example incidence of cancer during the year 2014, might be estimated by following up 4000 people for the full year, this would give you 4000 person years and then you would count the number of cases of cancer the divide it by 4000. This will give

you the incidence rate for 4000 person years. You would not want to compare this to the number of incidence in a group of 40,000 followed for the same about of time, nor would you want to compare this to be group of 4,000 people followed for 2 years. Clearly you would expected the numbers to be higher because there are more person years than 4,000. This is a summary statistic that is often given in Epidemiology without making a comparison, the general formula for a rate is.

The Rate = number of events/person units of time

If like above the units are years, this will give the estimated rate for a person year, but you can also use the same data to get estimated rate for a person day or month or minute, provided your data have are measured at that fine a level. Some examples of this are:

The Incidence Rate = number of new events/person units of time

The mortality Rate = number of deaths/person units of time

examples

THE SAMPLE MEAN OF THE EXAMPLE VARIABLES

- There is no meaningful sample mean z
- The sample mean of y is $2/6 = 1/3 = 0.333$.
  this is the proportion of 1s
- The sample mean of x is 7.45,
  which is the sum of $(0+0+1+2+3+4+5+6+9+12+40)/11$
- The sample mean of q is 8.22733,
  which is again the sum of the values divided by the sample size, n,
  which is again 11.

THE SAMPLE VARIANCE OF THE EXAMPLE VARIABLES

- There is no meaningful sample variance of z
- The sample variance of y is 0.2666667.
- The sample variance of x is 130.4727,
  $[(0-7.45)^2 + \ldots + (40-7.45)^2]/10$
- The sample variance of q is 153.409

FOCUSING ON X

median = 50th percentile = 2nd quartile = 4 (central position)

1st quartile = 25th percentile = 1 (median of values below median)

3rd quartile = 75th percentile = 9 (median of values above median)

min = 0th percentile = 0

max = 100th percentile = 40
range = max - min = 40
interquartile range = (3rd - 1st quartile) = 9 - 1 = 8

Summary Statistics For Associations

two binary or categorical variables

TWO BY TWO TABLES are a way to display two binary variables.

Consider a data sample such that we have two indicators sampled from a group of 7 people selected at a metro station near KI at 8 on a Monday morning in November. We ask two questions, 'Do you have a PhD?' and 'Does one of your parents have a PhD?' These are the data. having a PhD $= \{1, 0, 1, 0, 0, 0, 1\}$ and Parent with PhD $= \{0, 1, 1, 0, 0, 1, 1\}$

	Parent PhD	
PhD	0	1
0	2	2
1	1	2

This table displays the data, but is there a summary statistic that we can provide for the relationship between these two variables in our data? [13]

[13] YES THERE IS

RISK is a term used for the proportion of an outcome, such as having a PhD, generally within a given level of a different variables, like your parents having a PhD. Notation will be helpful here!

LET: The variable having a PhD be denoted as y and your parents having one be denoted by x. The proportion of people sampled with a PhD among those with parents that also have a PhD, the risk of having a PhD among those with parents that also have a PhD, will be denoted as $\widehat{p}_{phd}$ and is an estimate of $P(y = 1|x = 1)$.

Similarly, the proportion of people sampled with a PhD among those with parents that do not have a PhD, denoted as $\widehat{p}_{Nphd}$, is an estimate of $P(y = 1|x = 0)$ as is often called the risk of having a PhD among those with parents that do not have PhDs.

The Risk difference $= \widehat{p}_{phd} - \widehat{p}_{Nphd} = 2/4 - 1/3 = 0.1666667$

The Relative Risk or Risk Ratio $= \widehat{p}_{phd}/\widehat{p}_{Nphd} = (2/4)/(1/3) = 1.5$

[14]

[14] Although we have performed no inference, both summaries are in the direction that a higher proportion of people with parents that have PhDs are at risk of having a PhD themselves compared to people whose parents do not have PhDs.

ODDS is a term used for the ratio of those having and not having an

outcome, such as having a PhD or not having a PhD. This is different from the risk in that you are not dividing by the total!

$\widehat{odds}_{phd} =$
(number with PhD| Parents with PhD)/(number without PhD| Parents with PhD)
=2/2

$\widehat{odds}_{Nphd} =$
(number with PhD| Parents without PhD)/(number without PhD| Parents without PhD)
$= 1/2$

The Odds Ratio $= \widehat{odds}_{phd}/\widehat{odds}_{Nphd} = 1/0.5 = 2$ [15]

[15] This estimate again suggests higher odds of a PhD among those with parents with PhDs.

one binary or categorical variable and one numeric variable

CONDITIONAL MEANS, just like the proportion among those with subjects with a particular characteristic, are simply the sample means within a subgroup of the data defined by a different variable. Differences in the conditional sample means are most often used in this case as summary statistics.

Consider that among the 7 people selected for questioning we also ask "how old are you in years?" These are the data.

age $= \{38, 22, 35, 24, 18, 26, 54\}$
having a PhD $= \{1, 0, 1, 0, 0, 0, 1\}$ and
Parent with PhD $= \{0, 1, 1, 0, 0, 1, 1\}$

Sample mean of age|subjects with a Phd

$$= \frac{1}{3}(38 + 35 + 54) = 42.33333$$

Sample mean of age|subjects without a Phd

$$= \frac{1}{4}(22 + 24 + 18 + 26) = 22.5$$

DIFFERENCES IN CONDITIONAL MEANS is often used as a single summary of this association.

Sample mean of age|subjects with a Phd − Sample mean of age|subjects without a Phd

$= 19.83333$ [16]

[16] This is in the direction of subjects with the PhDs being older.

DIFFERENCE AND RATIOS OF RATES above we describe rates as a summary statistic that combines both a binary variable and an amount of time. Once the rate is calculated it is numeric, and if you do this is two different groups, the groups indicator is binary. Provided that both rates use the same unites. One can consider two summaries of association the difference in rates and the rate ratio.

The difference is rates = Rate per unit in group a − Rate per unit in group b

The Rate Ratio = Rate per unit in group a/Rate per unit in group b

Due to the models that are often used for rates, Rate Ratio is more common.

two numeric variable

CORRELATION is a measurement of trending together. Let us consider the x and q variables from above.

The linear correlation often call the Pearson's method: [17]

[17] This is a very high correlation, suggesting a linear association between x and q

$$\hat{r}_{xq} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(q_i - \bar{q})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(q_i - \bar{q})^2}} = 0.9985431$$

If you don't believe there is a linear relationship between the two numeric variables, then consider using the rank-based correlation as a summary of the association. We first need to rank the variables, this is called Spearman's method.

$R_x = \text{Rank}(x) = \text{order}(x) = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$
$R_q = \text{Rank}(q) = \text{order}(q) = \{1, 3, 2, 4, 5, 6, 7, 8, 9, 10, 11\}$

$$\hat{r}_{R_x R_q} = \frac{\sum_{i=1}^{n}(R_{xi} - \bar{R}_x)(R_{qi} - \bar{R}_q)}{\sqrt{\sum_{i=1}^{n}(R_{qi} - \bar{R}_x)^2}\sqrt{\sum_{i=1}^{n}(R_{qi} - \bar{R}_q)^2}} = 0.9840572$$

Displays Of Data

DATA VISUALIZATION is an important step in data analysis in order to understand how the data are structured and if there are errors in the data. It is also important for effectively communicating results of analyses to your peers. Data visualization, which includes tabular displays, is generally both the first and last step of a data analysis. The first step is to summarize the data and to understand the variables of interest. The last step is to effectively communicate your results and the main findings of your analysis. Therefore, in the last step it is critical to spend time designing your visualizations to improve understanding, to clearly and accurately communicate the quantitative information of interest.

GENERAL PRINCIPLES

- Encode quantitative data as linear lengths or distances. They are easier and more accurate to interpret than angles, area, volume, or color.
- Place things that are meant to be compared next to each other.
- Show the raw data.
- Remove extraneous information. Don't waste ink.
- Be aware that 5-10% of the population is color-blind so choose appropriate color schemes.

TABLES The following example illustrates the general principles as applied to tables. Some special considerations include the numerical precision (don't use too much precision), alignment of text and numbers, and avoiding clutter like colors and line separators.

	Parent PhD	No parent PhD	Difference
PhD	0.50	0.33	0.17
Age	34.5	26.7	7.8

FIGURES

single numeric variables

Show the raw data with a dot plot or strip chart. Show distributions with histograms or density plots. Show summary statistics with boxplots. Better yet, show the raw data and summary statistics in a boxplot with an overlayed scatterplot.

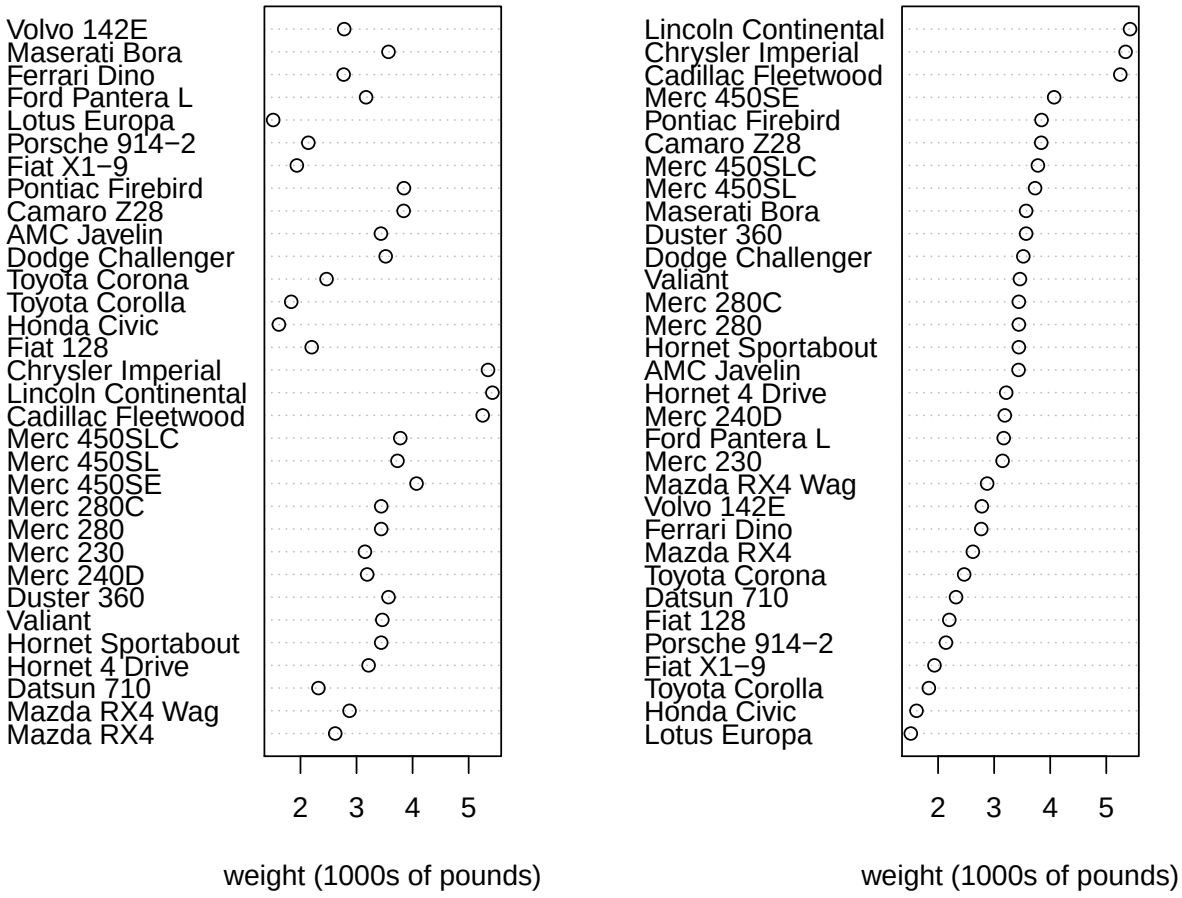


Figure 1: A dotplot shows the individual numeric values as the length along the x-axis. Unless the entries have some inherent ordering, then it is preferable to sort according to the numeric variable.

single categorical or binary variables

Pie charts are a bad way of displaying quantitative information. Humans are good at judging linear measures and bad at judging relative areas or angles. A bar chart or dot chart is a preferable way of displaying this type of data. [18] This statement is based on the empirical investigations of Cleveland and McGill as well as investigations by perceptual psychologists.

[18] Cleveland (1985), page 264: "Data that can be shown by pie charts always can be shown by a dot chart. This means that judgments of position along a common scale can be made instead of the less accurate angle judgments.

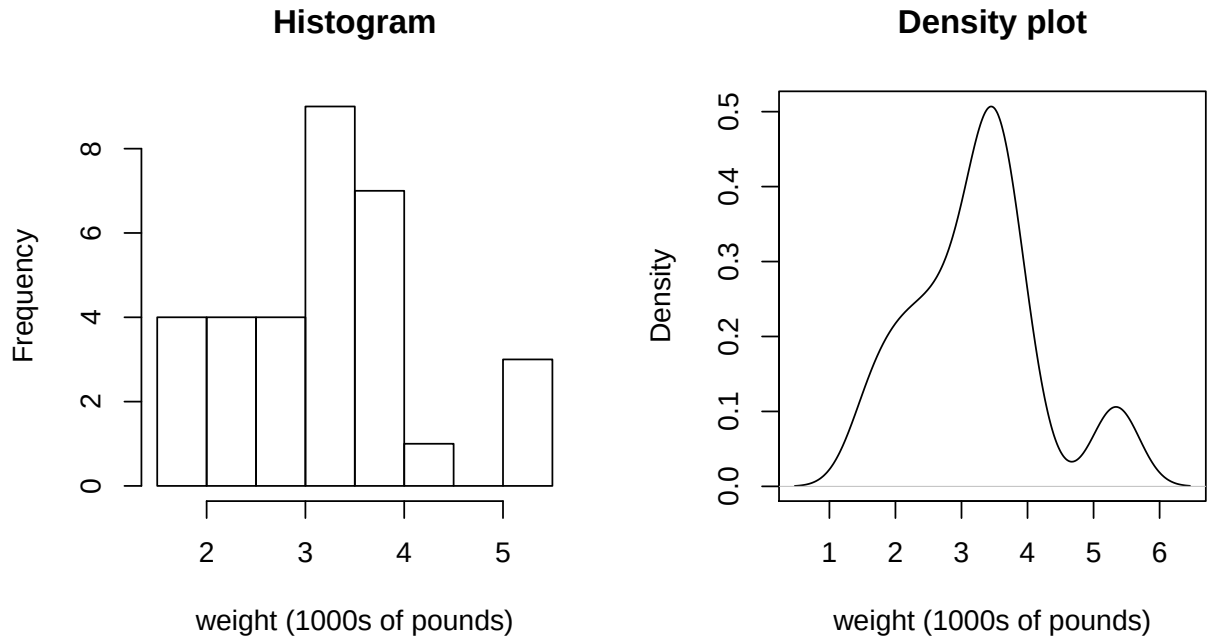


Figure 2: Histograms or Density plots for single numeric variables

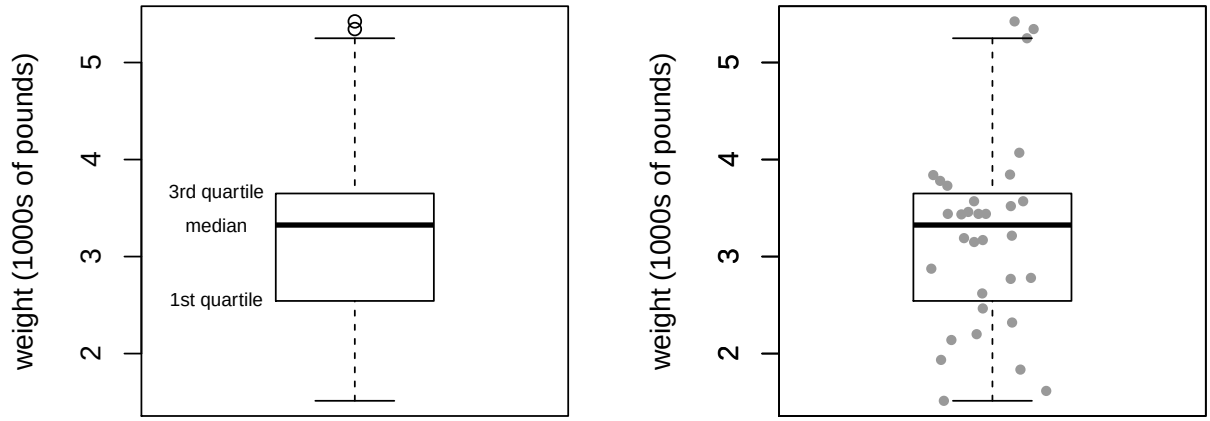


Figure 3: A marginal boxplot of a numeric variable will give you summary statistics of the variable. For small to medium datasets, it is good to overlay the raw data points on the boxplot.

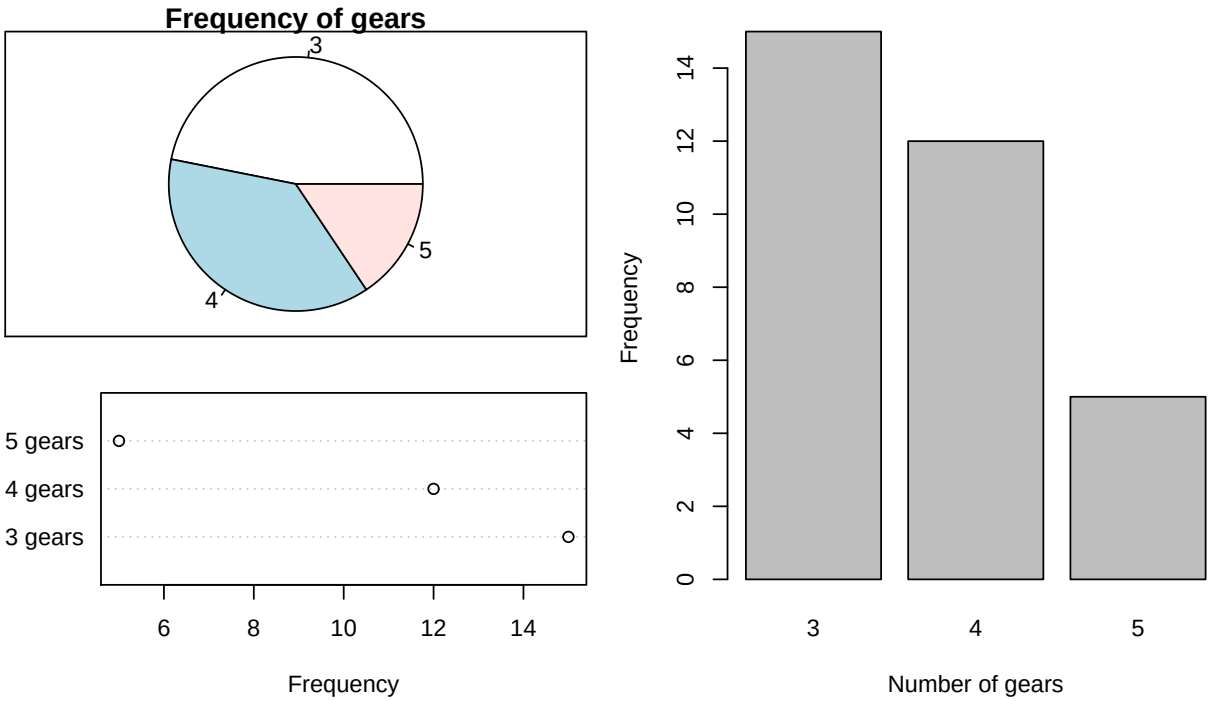


Figure 4: Pie charts show frequencies or percentages of categorical data as the areas of a circle. Dot charts or bar charts are preferred methods of visualizing this sort of information. The axes of the bar charts start at 0, whereas the dot charts do not necessarily.

two variables

With two or more variables, the focus turns to displaying the associations between the variables of interest. The same principles apply: encode numeric data as lengths rather than sizes or angles, don't waste ink, and show the raw data.

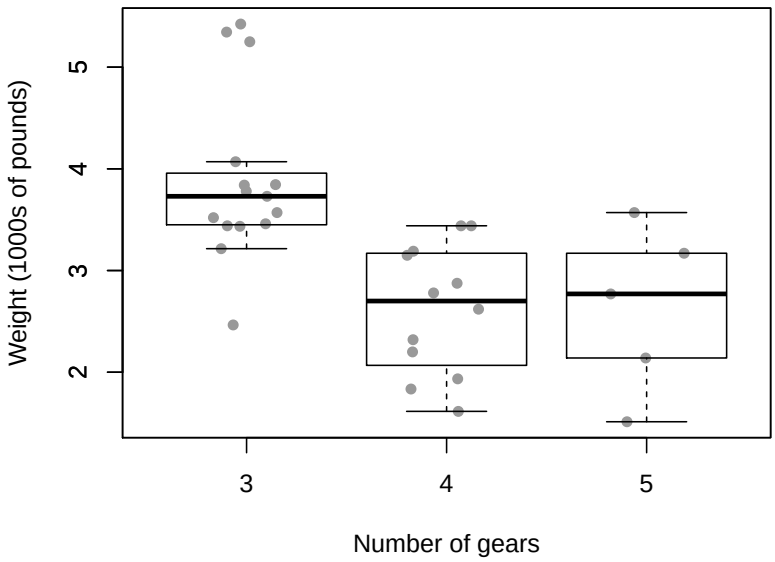


Figure 5: Boxplots naturally extend to displaying the association between a numeric variable and a categorical variable.

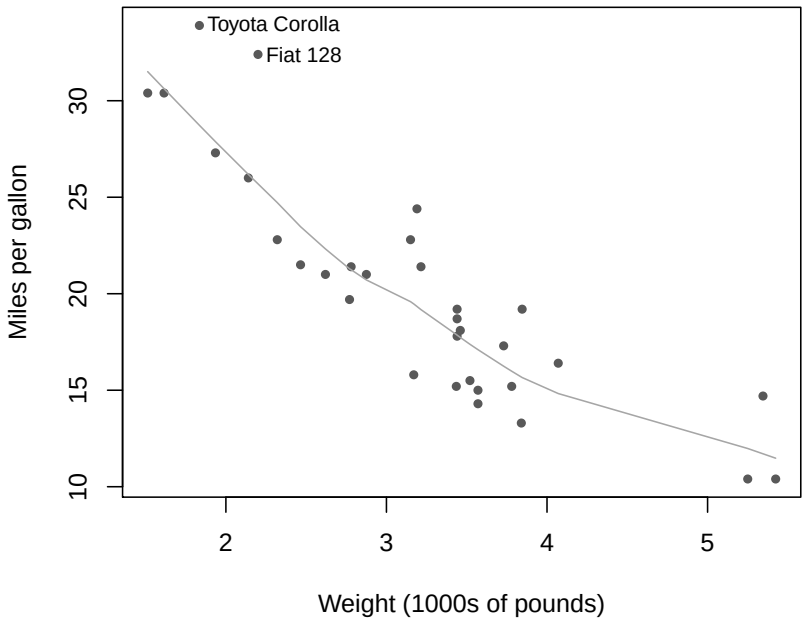


Figure 6: Scatterplots show the association between two numeric variables. Smoothing or regression functions can be overlayed to show summary statistics. Individual points or outliers can be highlighted with direct labels.

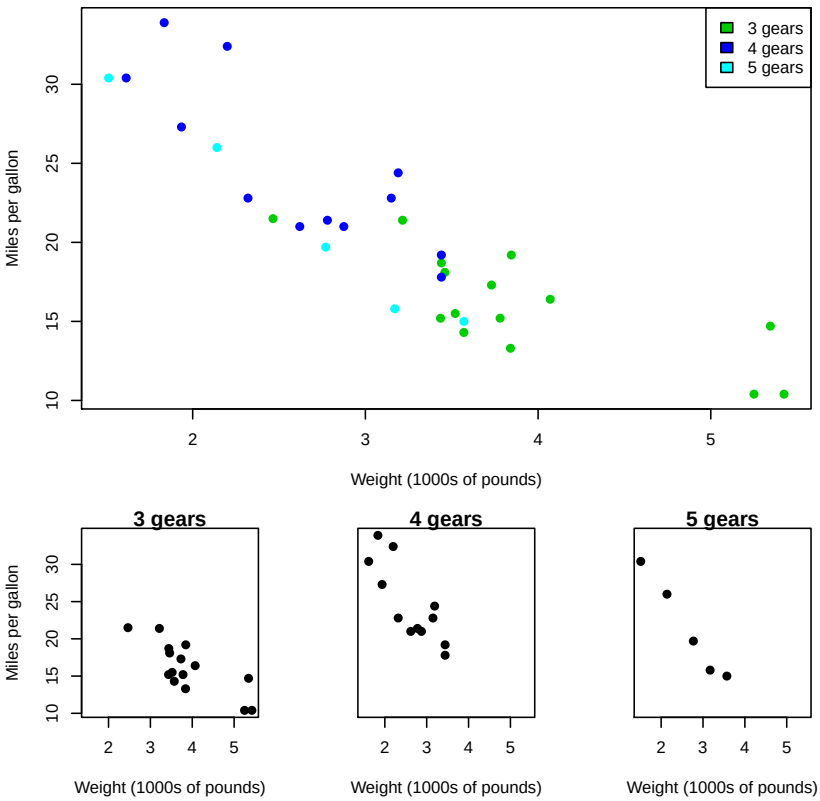


Figure 7: Three or more variables can be displayed on a plot in many ways. A third categorical variable can be encoding as color or shape, or used to create subplots in separate panels. Get creative, but remember the general principles. Consider the main message of the visualization, and the cognitive burden of matching colors or shapes to information. Three dimensional figures should be avoided.

Probability

Basic concepts

- $\mathrm{P}(A)$ is the probability of an event 'A'
- $\mathrm{P}(A)$ is between 0 (A is impossible) and 1 (A is sure)
- $\mathrm{P}(\text{ not } A) = 1 - \mathrm{P}(A)$

Joint probability:

- $\mathrm{P}(A \text{ and } B) = \mathrm{P}(A \cap B)$. This is equal to $\mathrm{P}(A) * \mathrm{P}(B)$, if A and B are independent events.
- $\mathrm{P}(A \text{ or } B) = \mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \text{ and } B)$. This is equal to $\mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A) * \mathrm{P}(B)$ if A and B are independent events. [19]

Conditional probability:

- $\mathrm{P}(A|B)$ = probability of A, given B. This is given by $\mathrm{P}(A \mid B) = \mathrm{P}(A \text{ and } B)/\mathrm{P}(B)$. [20]

Bayes' theorem:

connects $\mathrm{P}(A|B)$ to $\mathrm{P}(B|A)$ with the formula:

$$\mathrm{P}(A|B) = \mathrm{P}(B|A) * \mathrm{P}(A)/\mathrm{P}(B)$$

To see this, note that, using the formula for conditioning, we have

$$\mathrm{P}(A \cap B) = \mathrm{P}(A|B) * \mathrm{P}(B) = \mathrm{P}(B|A) * \mathrm{P}(A).$$

Example

Example 1: coin

$\mathrm{P}(T, H, T, T, H, T) = \mathrm{P}(T) * \mathrm{P}(H) * \mathrm{P}(T) * \mathrm{P}(T) * \mathrm{P}(H) * \mathrm{P}(T) = (0.5)^6$ (coin has no memory, i.e., results are independent)

[19] $\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B)$ if A and B are mutually exclusive, i.e., if $\mathrm{P}(A \text{ and } B) = 0$. Note that 'mutually exclusive' does not mean 'independent'! Actually, if A and B cannot happen together, it means that they are **not independent!**

[20] $\mathrm{P}(A|B) = \mathrm{P}(A)$ if $\mathrm{P}(A \text{ and } B) = \mathrm{P}(A) * \mathrm{P}(B)$, i.e., if A and B are independent. Knowing that I love pizza does not affect my prediction about tomorrow's weather. We can use this to determine what $\mathrm{P}(A \cap B)$ is: if we know $\mathrm{P}(A|B)$ then $\mathrm{P}(A|B) * P(B) = \mathrm{P}(A \cap B)$.

EXAMPLE 2: DICE

P(x even and x < 4) $= 1/6$ (only the value $x = 2$ satisfies both conditions).

Note that the result is NOT P(x even) $*$ P($x < 4$) $= (1/2) * (1/2) = 1/4$, because the two events are NOT independent!

EXAMPLE 3: DICE

P(1 or 2 or 5) $=$ P(1) $+$ P(2) $+$ P(3) $= 3/6 = 1/2$ (mutually exclusive events).

EXAMPLE 4: DICE

P(x even or $x > 3$) $=$ P(2 or 4 or 5 or 6) $= 4/6$, which is not equal to P(x even) $+$ P($x > 3$) $= 1/2 + 1/2 = 1$, because 'x even' and $x > 3$ are not mutually exclusive. The right calculation is P(x even or $x > 3$) $=$ P(x even) $+$ P($x > 3$) $-$ P(x even and $x > 3$) $= 1/2 + 1/2 - 2/6 = 4/6$, but $4/6$ is not equal to P(x even) $*$ P($x > 3$) $= 1/4$ is NOT equal to $2/6$. The values 4 and 6 are the ones for which both of these are true, which gives a probability of $2/6$ for the intersection (2 values out of 6). It can be seen when we look at the conditional that these are not independent.

EXAMPLE 5: DICE

P(x even $|x > 3$) $= 2/3$. This is because, if $x > 3$, then x is either 4, or 5, or 6, and 4 and 6 are even. This is a conditional probability. Using the formula: P(x even $|x > 3$) $=$ P(x even and $x > 3$)/P($x > 3$) $= (2/6)/(1/2) = 2/3$.

EXAMPLE 6: TWO DICE

P($x1 > 4|x2 = 3$) $=$ P($x1 > 4$) $= 2/6$, because the two dice are independent, and so the fact that $x2 = 3$ does not affect the chance that $x1 > 4$. Using the formula, P($x1 > 4|x2 = 3$) $=$ P($x1 > 4$ and $x2 = 3$)/P($x2 = 3$), but because the dice are independent, P($x1 > 4$ and $x2 = 3$) $=$ P($x1 > 4$) $*$ P($x2 = 3$), which simplifies the equation.

Random Variables and Distributions

A RANDOM VARIABLE is described by a set of possible values (the 'support' of the variable) and a function $f(x)$ that describes the associated probability.

FOR DISCRETE VARIABLES, the function $f(x)$ is called 'probability mass function' and directly returns the probability. This means that, for example, $\mathrm{P}(x = 4)$ is obtained by $f(4)$, i.e., by replacing $x = 4$ into $f(x)$. The sum of all probabilities is one, as it represents the chance of any event.

FOR CONTINUOUS VARIABLES, $f(x)$ is called 'probability density function' and probabilities are computed by integrating over $f(x)$, i.e., by measuring the area below the curve. By consequence, the probability of a specific value (e.g., $\mathrm{P}(x = 3)$) is zero for continuous variables because the area between 3 and 3 is zero. The total area is one, as it represents the chance of any event.

ALL RANDOM VARIABLES have a cumulative distribution function, which gives the probability of being less than or equal to a value, v, $\mathrm{P}(y \leq v)$. For discrete random variables this is obtained by summing the pmf, for continuous random variables it is obtained be integrating over the pdf.

Discrete random variables

A coin produces a binary random variable which follows a distribution called a Bernoulli, $\mathrm{Ber}(p)$. The parameter 'p' is the probability of success and is the parameter that defines this distribution.

BERNOULLI RANDOM VARIABLE $\mathrm{BER}(p)$:

- Possible results (support): $0, 1$.

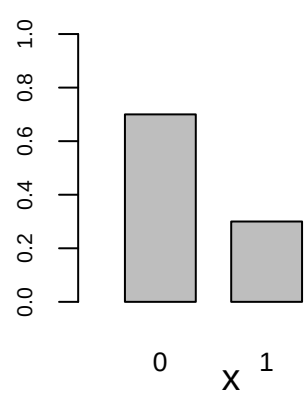


Figure 8: The pmf of a Bernoulli 0.3, the y-axis is the probability of the x value on the x-axis. This is directly given by the Bernoulli pmf

- In one sentence: $x \sim$ Ber(p), which you read as 'x has a Bernoulli distribution with parameter p'.
- The probability mass function is $f(x) = p^x(1-p)^{(1-x)}$ [21]
- The expectation of a Bernoulli random variable is p, by the formula $\sum x * P(x) = 0 * (1-p) + 1 * p = p$
- The variance x is $p * (1-p)$, by the formula $\sum x * P(x)^2 - p^2$

[21] which is a complicated way to say that $\mathrm{P}(x = 0) = 1 - p$, and $\mathrm{P}(x = 1) = p$.

Dice produce a discrete random variable which follows a distribution called discrete uniform, U(N). The parameter 'N' is the number of faces, each of which have the same probability and is the parameter that defines this distribution.

[22] there is also a continuous Uniform, which can be thought of as a 'die with infinite faces' (a sphere) is very important in probability theory, and can be used to generate random numbers from other variables

DISCRETE UNIFORM RANDOM VARIABLE U(N): [22]

- Possible results (support): $\{1, \ldots, \mathrm{N}\}$
- In one sentence: $x \sim$ U(N), i.e., 'x has a Uniform distribution with parameter N'.
- The probability mass function is $f(x) = 1/\mathrm{N}$
- The expectation of a discrete uniform random variable is $1 + \mathrm{N}/2$, or the sum of the ends of interval divided by 2.
- The variance is $\frac{(\mathrm{N}-1+1)^2-1}{12}$

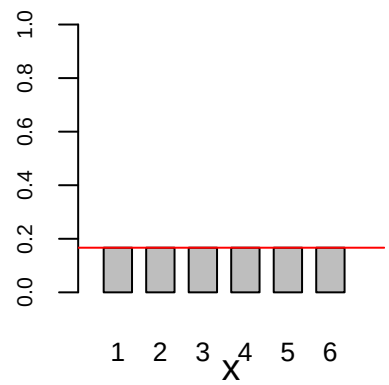


Figure 9: The pmf of a discrete uniform 6. the y-axis is the probability of the x value on the x-axis. This is directly given by the Uniform pmf, which is the same value for all values, here it is /6 and is indicated by the red line.

Counting the number of heads in n flips produces a discrete random variable which follows a distribution called Binomial, Bin(n,p). The parameter 'n' is the number of flips, and 'p' is the probability of success in each flip. A coin has no memory as we discussed in Chapter 3, so each flip is a new try, and the probability of success in two flips for example is $p * p$.

BINOMIAL RANDOM VARIABLE BIN(n, p):

- Possible results (support): $\{0, \ldots, n\}$
- In one sentence: $x \sim$ bin(n,p), i.e. 'x has a Binomial distribution with parameters n and p.'
- The probability mass function is $f(x) = n!/(x!(n-x)!) * p^x * (1-p)^{(n-x)}$ [23]
- The expectation of a binomial random variable is $n * p$
- The variance is $n * p * (1-p)$

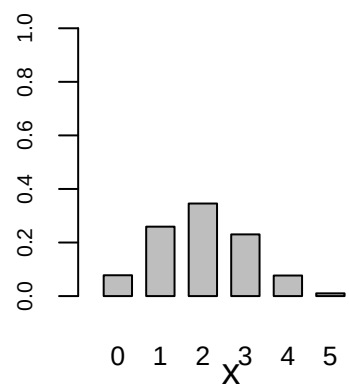


Figure 10: The pmf of a Binomial 5, p=0.4. The y-axis is the probability of the x value on the x-axis. This is directly given by the pmf.

[23] The notation 'x!' means $x * (x\breve{}1) * (x\breve{}2) * \ldots$ and is read 'factorial of x'. For example, 5! = 5*4*3*2*1 = 120.

The number of events in a given space or time, when the maximum number of events (the 'n' of the binomial) is not known or very large is a random variable with a Poisson distribution, Pois(λ). The parameter 'λ' is the average number of events in the given space/time and defines the distribution.

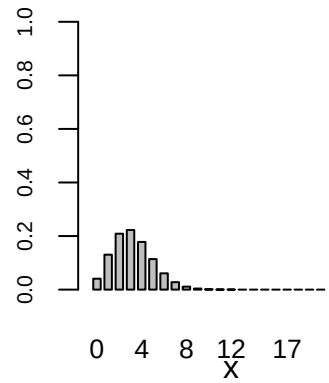


Figure 11: The pmf of a Poisson 3.2. The y-axis is the probability of the x value on the x-axis. This is directly given by the pmf. The possible values actually continue up to ∞, but the probabilities become very very small after around 10.

POISSON RANDOM VARIABLE POIS(λ):

- Possible results (support): $\{0, 1, 2, \ldots, \infty\}$
- In one sentence: $x \sim$ Pois(λ), i.e. 'x has a Poisson distribution with parameter λ.'
- Probability mass function: $f(x) = e^{-\lambda}\lambda^x/x!$ [24]
- The expectation is λ [25]
- The variance is λ

Other discrete distributions exist, Google it! But the above are the distributions you will need to know by heart in general for Epidemiology research.

[24] $e \approx 2.71$ is Euler's number
[25] Because the expectation and the variance are the same, this is sometimes considered a very restrictive distribution

Continuous random variables

LIKE THE DISCRETE UNIFORM, THERE IS A CONTINUOUS UNIFORM, which can be thought of as a die with infinite faces, a perfect sphere. If you roll it on a perfectly smooth surface that probability of it landing on a single value is 0.[26] Rolling it on a not exactly smooth surface you might get a range of values touching the surface when it stops, and there is a probability of that range. This is one way to connect the easy to think about discrete random variables to the continuous. Start with a 6 sided die, and keep increasing the flats by filing off points until you have a sphere.
A random variables that follows a continuous uniform distribution, has parameters a, the lowest value, and b the heights value which define the distribution.

[26] actually it would probably never stop

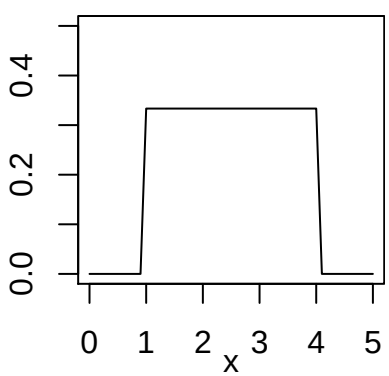


Figure 12: The pdf of a continuous uniform from 1 to 4.

CONTINUOUS UNIFORM RANDOM VARIABLE UNIF(a, b):

- Possible values (support): $[a, b]$
- In one sentence: $x \sim$ Unif(a, b), i.e., 'x has continuous Uniform distribution from a to b'
- The probability density function is $f(x) = \frac{1}{(b-a)}$ if $x \in [a, b]$ and 0 otherwise.

- The expectation is $(a+b)/2$
- The variance is $\frac{(b-a)^2}{12}$

The time between events, when the number of events during a period of time follows a Poisson distribution, is a random variable which follows a Exponential distribution, $\text{Exp}(\lambda)$. The parameter λ defines the distribution.

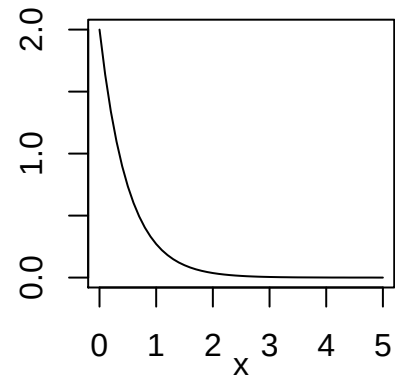


Figure 13: The pdf of an exponential 2.

Exponential random variable $\text{Exp}(\lambda)$:

- Possible values (support): $[0, \infty)$
- In one sentence: $x \sim \text{Exp}(\lambda)$, i.e., 'x has an Exponential distribution with parameter λ'
- The probability density function is $f(x) = \lambda e^{-x\lambda}$
- The expectation is $1/\lambda$
- The variance is $1/\lambda^2$

The heights of all the people working at Karolinska is a random variable with a normal or Gaussian distribution. The parameter μ is the mean, and the parameter σ^2 is the variance, and these define the distribution. [27]

[27] When $\mu = 0$ and $\sigma^2 = 1$, this is often referred to as the standard normal.

Normal or Gaussian Random Variable $\text{N}(\mu, \sigma^2)$:

- Possible results (support): $(-\infty, \infty)$
- In one sentence: $x \sim \text{N}(\mu, \sigma^2)$, i.e., 'x has a Normal distribution with parameters μ and σ^2'
- The probability density function is $f(x) = \frac{1}{\sqrt{\sigma^2 \pi}} e^{(x-\mu)^2/\sigma^2}$.
- The expectation is μ
- The Variance is σ^2

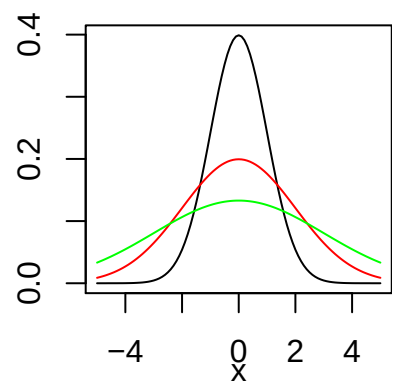


Figure 14: The pdf of a normal with mean 0, and standard deviation 1, in black, 2 in red and 3 in green.

There are not a lot of easy examples of random variables that are distributed T. The Student's T is a variation of the standard Normal. The parameter ν is the defining parameter of the student's t distribution and it called the degrees of freedom. [28].

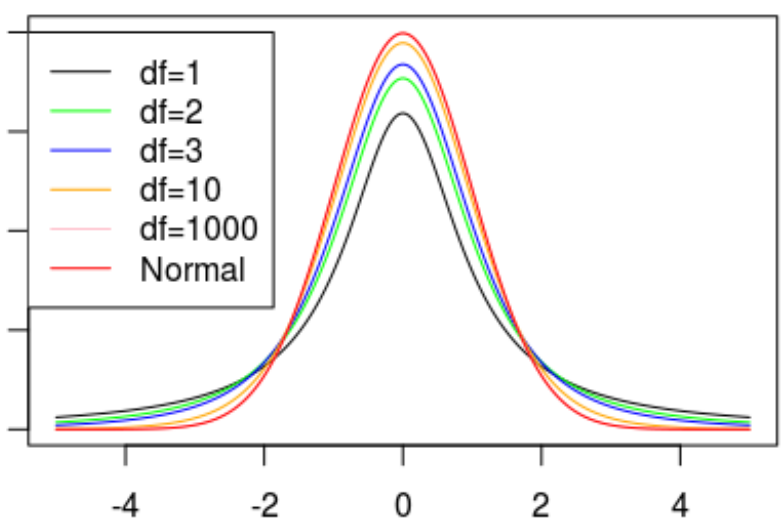
[28] we will use the t distribution for inference in later chapters

STUDENT'S T RANDOM VARIABLE $t(df)$:

- Possible values (support): $(-\infty, \infty)$
- In one sentence: $x \sim t(\nu)$ i.e., 'x has a Student's t distribution with degrees of freedom ν'.
- The probability density function is $f(t) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi}\,\Gamma(\frac{\nu}{2})}\left(1+\frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}$, [29]
- The expectation is 0, if $\nu > 0$ and undefined otherwise.
- The Variance is $\nu/(\nu-2)$ if $\nu > 2$, ∞ for $1 < \nu \leq 2$ and undefined otherwise.

[29] Where $\Gamma(z) = \int_0^\infty x^{z-1}e^{-x}$ is the Gamma function or Euler's integral of the second kind. You DO NOT need to know this!

Figure 15: As $\nu \to \infty$, the T goes to a standard normal.



CALCULATING PROBABILITY FOR CONTINUOUS RANDOM VARIABLES is not as easy as for discrete variables. We need to determine the area under the pdf which is the integral of the pdf $\int_a^b f(x)dx =$

P$(a < x < b)$, sometimes this has a closed form and sometimes it does not. For all proper distributions, the area under the pdf over the full support is 1, this is the probability of the event being in the support, and this is of course 1, $\int_{support} f(x)dx = 1$. Therefore the probability of x being between a and b, P$(a < x < b)$, is the area under the pdf contained between a and b, as compared to the whole, 1, this is the proportion of the area.

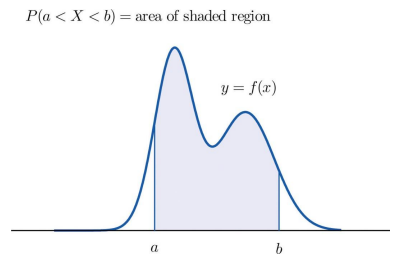
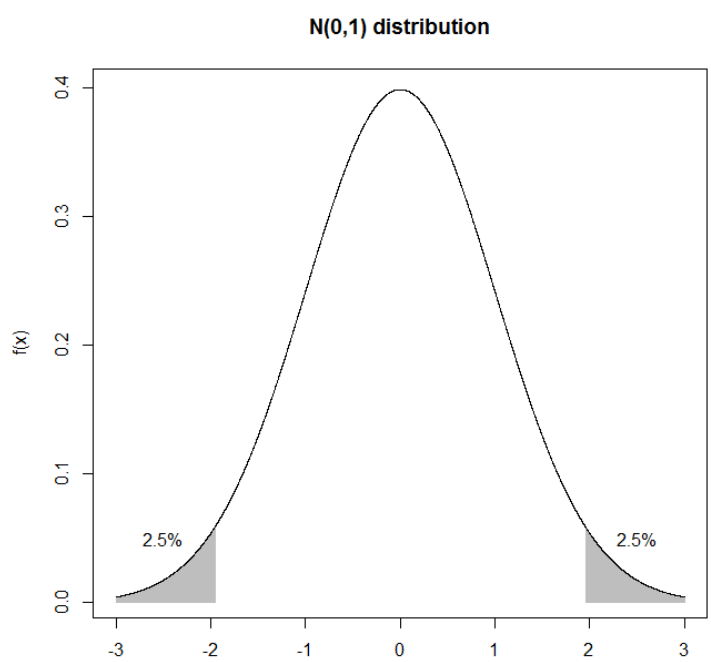


Figure 16: Calculating the probability of a set of responses requires calculating the area

Figure 17: Tail area probabilities for the standard normal distribution.

Standard Normal distribution



THE STANDARD NORMAL is a special case of the normal with expectation $\mu = 0$ and variance $\sigma^2 = 1$. It is a very useful distribution because the probabilities have been worked for almost all meaningful values, and all normal random variables can be easily converted to it. [30] The Normal distribution is symmetric about μ regardless of μ or σ^2, because of this, if x follows a standard normal then the P$(x > 1.96)$ is the same as P$(x < -1.96)$, and this probability represents the proportion of the area under the curve contained between 1.96 and ∞ or between -1.96 and $-\infty$. For a standard normal this is 0.025. .5cm[0]

[30] We will also see in the next chapter that because of the CLT this is the most useful distribution for inference.

[0] This is an important cut-point because someone one time decided that a probability of 0.05 or less is meaningful (p-value <0.05 is often called significant)

To compute probabilities for a standard normal we again need to compute areas. For example, if I tell you that $x \sim$ N(2,9), then P($x > 8$) would be computed as the area of the corresponding probability density function, $f(x)$, between 8 and infinity.

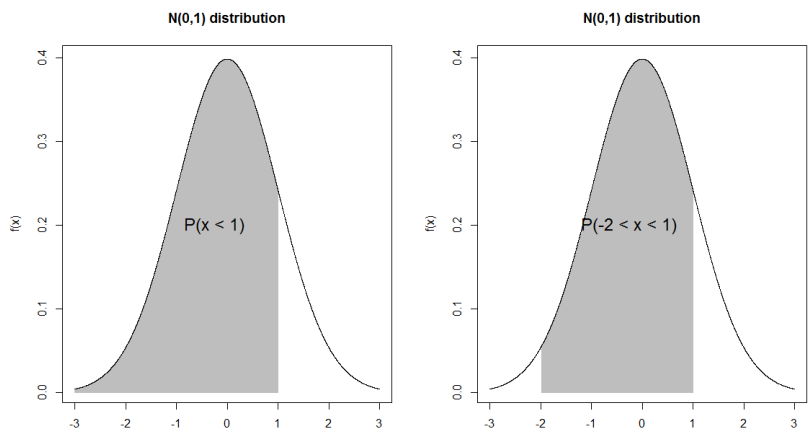


Figure 18: Computing probabilities as areas under the normal distribution.

At the end of a textbook you typically find the 'tables of the normal distribution'. The tables have been computed for a specific Normal distribution: the N$(0, 1)$, which we call the 'standard Normal'. There is nothing special in the fact of having mean = 0 and variance = 1.

However, any normal can be converted into the standard normal as follows: take the variable, subtract the mean, and divide by the standard deviation. In other words, if $x \sim$ N(μ, σ^2), then the standardized variable

$$z = (x - \mu)/\sigma$$

has a standard normal distribution, i.e., $z \sim N(0, 1)$. [31]

[31] Usually, we don't know what μ actually is or σ, we need to estimate them. Ideas on how to do that???

When we know x is normal this produces a standard normal, however any data can be standardized. Take the values $x = \{1, 2, 3\}$. the sample mean $\overline{x} = 2$ and the sample standard deviation is 0.8165. We can use these to standardize the data. You can easily check that the mean is 2, and the standard deviation is about 0.8165.

If you compute the new values:

$$z = (x - 2)/0.8165 = \{-1.2247, 0, 1.2247\},$$

you will see that the mean z is zero, and the variance of z is 1. This operation is called standardization and the standardized variable will have a standard Normal distribution when the original variable is normally distributed, i.e., $z \sim N(0, 1)$ when $x \sim$ N.

CONSIDER AGAIN THE VARIABLE $x \sim \mathrm{N}(2, 9)$, and suppose you want to calculate $\mathrm{P}(x > 8)$.

$$\mathrm{P}(x > 8) = \mathrm{P}((x - \mu)/\sigma > (8 - \mu)/\sigma)$$

$$\mathrm{P}(z > (8 - 2)/3) = \mathrm{P}(z > 2).$$

You converted the probability of a N(2,9) distribution into the probability of a N(0,1) distribution! You found that $\mathrm{P}(x > 8) = \mathrm{P}(z > 2)$, and since we know $\mathrm{P}(z > 2) = 0.02275013$, we know that $\mathrm{P}(x > 8) = 0.02275013$, or by a table 0.02.

Summary of random variables and distributions

THERE ARE RANDOM VARIABLES x, [32] from which we draw a sample of values, and these are governed by distributions which have particular characteristics. They all have parameters, which are used in the pdf to define the distribution. There are discrete and continuous distributions and above we outline a number of useful versions of both. The standard normal distribution is of particular interest for inference as we will see later.

[32] In textbooks, random variables are denoted with a capital letter (e.g., X), while the lowercase (e.g., x) is used to indicate a generic value of X. For example, if X is a die, $f(x)$ corresponds to $\mathrm{P}(X = x)$, e.g., $f(4) = \mathrm{P}(X = 4)$. To simplify the notation and avoid confusion, in these notes we use the lowercase x instead of X, and a specific numeric value (e.g., 4) instead of the generic x.

FOR A DISCRETE DISTRIBUTION the pmf $f(x)$ gives you the probability of x, for a continuous distribution the pdf $f(x)$ **is not a probability** and in fact if x has a continuous distribution $\mathrm{P}(x = c) = 0$ for any single value c. Instead, we need to consider a range of values and the area under the pdf, $\int_a^b f(x)dx$ to calculate probabilities for continuous random variables.

THE NORMAL distribution is very useful for inference as we will see in the next chapter, and you can convert all normal random variables to the standard normal, for which all the probabilities have already been worked out!

THE STUDENT'S T DISTRIBUTION is just a variation of the standard Normal distribution and looks almost identical to it with high enough degrees of freedom. The usage of the t and Normal distributions will be clarified in the next section.

ALL DISTRIBUTIONS, except the Normal and Student's t, have been given a simple interpretation:

Distribution	Example
Bernoulli, Ber(p)	A coin with P(x = 1) = p
Discrete Uniform, U(N)	A die with N faces
Binomial, Bin(n,p)	Number of successes in n trials, with P(success) = p
Poisson, Po(λ)	Number of events (earthquakes, accidents) in a given space and/or time, with λ = mean number of events
Continuous Uniform, Unif(a, b)	a horizontal line from a to b
Exponential, Exp(λ)	Time (or space) between two events (earthquakes, accidents), when λ = mean number of events
Normal N(μ, σ^2)	What has a Normal distribution? Maybe measurements of height that have not been rounded in any way...
Student's t, t(df)	What has a t distribution?

Table 1: Examples from each distribution

Name	Notation	Expectation	variance	Support	Type
Bernoulli	$x \sim$ Ber(p)	p	$p * (1-p)$	$\{0, 1\}$	Discrete
Uniform	$x \sim$ U(N,a,b)	$a + b/2$	$(b - a + 1)^2 - 1/12$	$\{a, \ldots, b\}$	Discrete
Binomial	$x \sim$ Bin(n,p)	$n * p$	$n * p * (1-p)$	$\{0, 1, \ldots, n\}$	Discrete
Poisson	$x \sim$ Pois()	λ	λ	$\{0, 1, \ldots, \infty\}$	Discrete
Continuous Uniform	$x \sim$ Unif(a, b)	$1/(a+b)$	$(b-a)^2/12$	$[a, b]$	continuous
Exponential	$x \sim$ Exp(λ)	$1/\lambda$	$(1/\lambda)^2$	$(0, \infty)$	continuous
Normal	$x \sim$ N(μ, σ^2)	μ	σ^2	$(-\infty, +\infty)$	continuous
Student's t	$x \sim$ t(df)	0 if df > 1	df/(df $-$ 2) if df > 2	$(-\infty, +\infty)$	continuous

Table 2: Parameters and Notation for the Distributions we have discussed

There are a lot of other distributions! Examples:

- Geometric, Hypergeometric, Negative Binomial (discrete);
- Uniform, Beta, Gamma, Weibull, Logistic (continuous).

Single Sample Hypothesis Testing and Inference for the Expectation

The concept of inference is a more general idea than statistics, and has to do with deducing things based on evidence. Statistical inference involves using data as evidence, and making conclusions (or not) about parameters of the data generating mechanism (DGM). In the absence of sufficient evidence, then we cannot always make a conclusion. Frequentist [33] inference is always based on the data but is done into to say something about the fixed data generating mechanism of the random variable. In a parametric sense, this is inference about the fixed expectation of the distribution that governs the random variable from which are sampling. You are attempting to use the things you can calculate about your sample to give you information about that expectation.

[33] there is also Bayesian and Likelihood based inference

From Chapter 1 we have a hypothesis about the DGM and we have collected data to test that hypothesis. We know how to summarize that data now, and we know some of the distributions the random variables that generated those data may follow. How do we test our hypothesis? How do we account for the fact that we sampled?

For example: If you collect the weights of $n = 10$ rats, and you compute the mean weight, you get a number, e.g., $\overline{x} = 350$ grams. If you repeat the experiment with another sample of 10 rats, you will obtain a different number e.g., $\overline{x} = 347$. Which of these numbers is correct? They are both estimates of the true DGM expectation of the distribution that govern rat weights. However, they are likely both wrong in the sense that none of them will be exactly equal to the true DGM expectation of the distribution that govern rat weights.

You can make a simple experiment with a coin (p = 0.5).

Flip 10 times

I did and got the following:
Experiment 1: HTTHHHTHTH, estimated P(T) = 0.4
Experiment 2: HHHTTTTHTT, estimated P(T) = 0.6

We know the true probability of tail was 0.5 in the underlying data generating distribution from which the sample was taken.[34] However, in a finite sample, the estimated probability will usually be different from the true value. The true data generating mechanism parameter can be thought of either as the distribution that generated the random variable from which we took a sample or the parameter from a infinitely large sample (rerunning the universe in the case of fixed populations). Either way this is what we are doing inference for, we are simply using our sample to do so.

[34] assuming the coin is fair

THE FACT THAT when you repeat the experiment, you obtain a different value of $\overline{x}$, introduces a fundamental concept: the sample mean is a random number. Just like x, $\overline{x}$, is governed by a distribution that determines its behavior. In general, whatever you compute from a sample will be random over repeated sampling.

For example, the sample median, the sample variance, the sample quantiles, will all be random. This is why we do inference: to characterize the intrinsic randomness in the summary statistics that we use to describe the data and attempt to say something about the DGM.

INFERENCE (REJECTION/NON-REJECTION) Inference is NEVER about the current data for this reason. If the p-value is 'small' it says something about the DGM or the super population, not the current data.[35]

[35] Now we understand what we are doing but how do we actually do it?

Inference for the expectation

BASED ON $\overline{x}$, CAN WE SAY ANYTHING ABOUT μ_x? Yes we can, because $\overline{x}$ has a lot of very nice properties and one super super cool theorem about it! The sample mean $\overline{x}$ is a random variable! Same as any random variable, it will have a distribution that governs the random variable from which it was generated.

The sample mean $\overline{x}$ has a variance, meaning the variation over all possible values it takes on, this is informed by the distribution of x, for our estimator $\frac{1}{n}\sum x_i$ the true variance is known to be σ_x^2 the true DGM variance of x divided by the sample size n. The quantity $\sigma_x/\sqrt{n}$ is the standard deviation of all possible values of $\overline{x}$ and is called the *standard error of the mean* and we can estimate it from the data. An estimator of that variance of the sample mean is $\widehat{\mathrm{Var}}(\overline{x}) =$

$\widehat{\mathrm{Var}}(x)/n$ and an estimator of the standard error of the mean or just the standard error is the square root of this.

The estimator $\frac{1}{n}\sum x_i$ has several useful properties.

UNBIASEDNESS OF THE SAMPLE MEAN is shown by taking the average of all possible values of your estimator, $\frac{1}{n}\sum x_i$, is equal to the true DGM expectation. the mean of all possible values of $\overline{x}$ is μ. This means that your specific $\overline{x}$ may be less or more than the true DGM μ, but 'on average', you are doing well. You say that $\frac{1}{n}\sum x_i$ is an 'unbiased estimator' of the true DGM μ. It is like having a target, and you are able to aim for the center with an unbiased estimator.

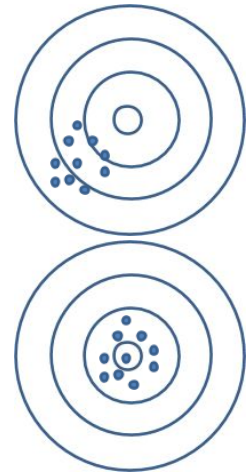


Figure 19: An Unbiased estimator of the expectation on the bottom and a biased estimator on the top

IF AN ESTIMATOR OF A PARAMETER approaches the true DGM value
of that parameter as n goes to infinity, then the estimator is said to be consistent. If the variance of all possible values of your estimator goes to zero as n goes to infinity and it is unbiased, it is consistent. Our estimator $\frac{1}{n}\sum x_i$, we know to have variance σ_x^2/n, then if $\sigma_x^2 < \infty$, this goes to zero as $n \to \infty$. This means that your specific $\overline{x}$ will have less variance, i.e., it will probably be closer to the true DGM μ, if n is larger.

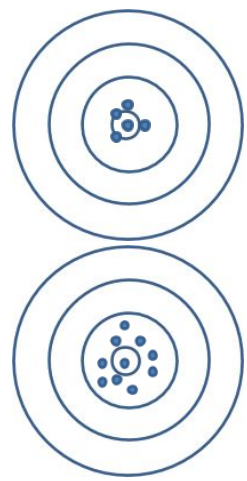


Figure 20: A consistent estimator of the expectation with a sample size of 1000 on the top and a sample size of 10 on the bottom

TO SUMMARIZE SO FAR:

- We have a sample of n observations from a random variable x with true DGM expectation μ_x and variance σ_x^2.
- We compute the sample mean by using the estimator $\frac{1}{n}\sum x_i$, the value of which we denote as $\overline{x}$
- The sample mean, $\overline{x}$, is random variable with a true DGM $\mu_{\overline{x}}$ and variance $\sigma_{\overline{x}}^2$
- We know that $\frac{1}{n}\sum x_i$ is unbiased, meaning $\mu_{\overline{x}} = \mu_x$ and that $\sigma_{\overline{x}}^2 = \sigma_x^2/n$.
- This means $\frac{1}{n}\sum x_i$ is consistent as well as unbiased, because $\sigma_x^2/n \to 0$ as $n \to \infty$, and therefore it must converge to a value and because it is unbiased that value is μ_x

We already know a lot about the random variable x, but there is another fundamental piece of information of inference.

The Central limit theorem

THE CENTRAL LIMIT THEOREM (FORMAL):

Let $\{x_1, \ldots, x_n\}$ be a random sample of size n that is a sequence of independent and identically distributed (i.i.d.) random variables drawn from a distribution with expected value given by μ and finite variance given by $\sigma^2 < \infty$, letting $\overline{x} = \frac{1}{n}\sum x_i$. Then as $n \to \infty$,

$$\sqrt{n}(\overline{x} - \mu) \to^d \mathrm{N}(0, \sigma^2).$$

Meaning that $\sqrt{n}(\overline{x} - \mu)$ converges in distribution to $\mathrm{N}(0, \sigma^2)$.

THE CENTRAL LIMIT THEOREM (MEANING): For large n, **regardless** [36], of the true DGM of the sample, the sample mean $\overline{x}$ will approach a normal with mean μ and variance σ^2/n. $\overline{x}$ is approximately $\sim N(\mu, \sigma^2/n)$. OR in large enough samples sizes if we standardize $\overline{x}$, we obtain a standard Normal variable, of which we know all properties.

[36] as long as the true DGM has a μ and finite variance given by $\sigma^2 < \infty$

$$z = \frac{(\overline{x} - \mu)}{\sigma/\sqrt{n}} \sim \mathrm{N}(0, 1)$$

THE CLT THEOREM TELLS us a property of the sample mean, $\overline{x}$. It is wonderful, because we know these properties from a THEOREM[37], i.e., before collecting the data! CLT is used to compute confidence intervals and perform hypothesis testing and compute p-values (in most cases). If you don't understand CLT, you will never understand p-values and confidence intervals. So, make sure everything is clear before proceeding. Using the CLT and the sample mean, we can derive the null distribution we should compare to, in order to obtain inference. Without the CLT we need to make more assumptions or do further analysis or sampling to know what the distribution under the null therefore we would not have a way of gain evidence against the null. Without the CLT, we could still determine the null if we knew that x was normal or t or exponential. [38]

[37] a theorem is a statement or result that can be proven mathematically to follow from the accepted conditions.

[38] There are ways to approximate or obtain a sample from the distribution under the null using the data, and we will discuss this in later chapters, where one simulates repeated sampling enforcing the null based on the data.

SIMILAR THEOREMS have been derived for many other statistics: the median, the difference between two means or medians, the hazard ratio and odds ratios, the correlation coefficients, the regression coefficients. Here you will only find details about the sample mean, but we will use other versions of CLT for hypothesis testing and inference on regression, correlation. These are referred to as the asymptotic properties of a test, what happens when the sample goes to infinity, and almost every test we discuss will rely on these asymptotic properties and our sample size being large enough for these properties to at least approximately hold. [39]

[39] This is sometimes called being in asymptotia

WE DON'T KNOW μ and that is the point. So, how do we use the CLT to test things? Well we can make a guess about μ, and see what we get. We know by the CLT that if our guess for μ, 2, is correct and assuming we know σ [40], that $z = \frac{(\overline{x}-2)}{\sigma/\sqrt{n}} \sim \mathrm{N}(0,1)$

Now if we are wrong, and $\mu = 8$ then $z \sim \mathrm{N}(6,1)$, not $\mathrm{N}(0,1)$ as expected. The as expected is the null to compare to, so assuming that $\mu = 2$, $z \sim \mathrm{N}(0,1)$. If we obtain a value that is in the tail of $\mathrm{N}(0,1)$ that makes us question if we guessed μ correctly. If z was very far in the tail in fact, we might reject our guess all together, thus **rejecting that null**. Now let's bring it all together!

[40] Seriously just ignore this part for a minute please!

IN CHAPTER 1 we stated an hypothesis, the true DGM expectation of rat weights is 134 grams. We got a sample of 1000 rats and weighted them and find the sample mean is 138, and let's say a fairy tells you that $\sigma_x = 2$, and we therefore know that the standard error of the mean is $2/\sqrt{1000} = 0.06324555$.

By the CLT, we know that if our guess of 134 grams is correct, then $(138 - 134)/0.06324555$ comes from a standard normal. As the support of a standard normal is $(-\infty, \infty)$, we can never say this isn't possible, but $(138 - 134)/0.06324555 = 63.24556$, and $\mathrm{P}(z >= 63.24556)$ [41] the probability that z is at least as large as what we have observed, **IF** z **IS STANDARD NORMAL**, is so so close to 0 that you might call it approaching impossibility. Because of this, I would be encouraged to believe my guess was wrong and that I had evidence to reject that guess and therefore the null based on what I have observed.

[41] $\mathrm{P}(z >= 63.24556)$ =$\mathrm{P}(z > 63.24556)$, because the standard normal is continuous.

Now say the fairy tells you that $\sigma_x = 200$, then our test statistic is $138 - 134/6.324555 = 0.6324556$, and then $\mathrm{P}(z >= 0.6324556)$, **IF** z **IS STANDARD NORMAL**, is 0.2635446. I am not encouraged to reject the null. **BUT I CANNOT ACCEPT THE NULL. THIS DOESN'T PROVE THE NULL**. There are many many other distributions this might have come from, just because it is not approaching impossibility for the standard normal does not mean it must come from the standard normal. This gives me no evidence to support the null, it just fails to give me sufficient evidence to reject it. [42]

[42] This is a very important and someone will make it in class until their face is blue, I promise.

Hypothesis testing via a one sample Z-tests

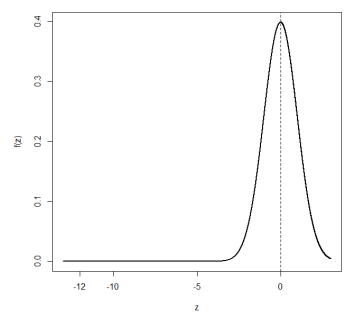
IF WE ARE IN ASYMPTOTIA then we don't need a fairy to tell us the DGM variance, we can estimate it, and the test statistic we were using above, with DGM standard deviation σ replaced by the sample standard deviation is still standard normal under the null. [43].

[43] We wave our hands over this because although it is an important point, it is not something you need to worry about. Either you have a large enough sample to use the CLT or you don't, the estimation of the standard error should not be something you worry about accounting for, unless you are in a small sample. If you are in a small sample, we need to use a different test anyway!

A ONE SAMPLE Z STATISTIC is simply the formula we have given above for standardization. the test statistic based on your data to test that your guess for μ_x, let's call it μ_g, g for guess, is $\hat{z} = \frac{(\overline{x}-\mu_g)}{\hat{\sigma}/\sqrt{n}}$. We calculate this by calculating the sample mean using the estimator $\overline{x} = \frac{1}{n}\sum x_i$, and the estimated standard deviation is 0.01. From this you obtain $\hat{z}$, and you know by the CLT that it should have a standard normal distribution if 16 is a large enough sample. [44] So, we can compare to that distribution to see the probability of values larger or smaller than $\hat{z}$ to see if we have evidence against the guess.

[44] which we will assume it is for this example.

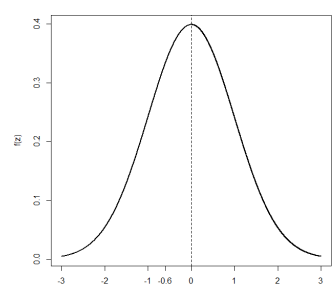
SUPPOSE YOU ARE BUYING WINE, and the seller tells you that, on average, each bottle contains 1 liter. To verify this, I sample n = 16 bottles. I find that the sample mean $\overline{x} = 0.97$, and that the sample standard deviation is 0.01. On average, there is less wine than expected. Is this sufficient evidence to reject the null and call the wine seller a cheater? Let's check! the Z-statistic here is $(0.97 - 1)/(0.01/4) = -12$. the $\mathrm{P}(z < -12)$ is $1.776482e-33$ or 2 with 0.(33 zeros) in front of it. It is really really small. Meaning that under the null, the probability of something smaller than -12 is very low. Because we are interested in the null that $\mu = 1$ liter, and not that $\mu \geq 1$ liter, for which we would only be interested in $\mathrm{P}(z < -12)$, we also want to consider that we got the direction wrong, $\mathrm{P}(z > 12)$. Because of the standard normal is symmetric, $\mathrm{P}(z > 12) = \mathrm{P}(z < -12)$, and so to consider both sides, we simply need to double, $1.776482e-33$.



MORE FORMALLY, the null hypothesis, often written H0: $\mu = 1$ liter. Our hypothesis is that the bottles contain something other than 1 liter, larger or smaller. [45].

[45] The hypothesis stated and what you want to provide evidence for, is not the same as the null hypothesis. In general the null will be the opposite!

NOW LET US consider that instead we obtain a sample with a sample mean of 0.9985, $(0.9985 - 1)/(0.01/4) = -0.6$ and $\mathrm{P}(z < -0.6) = 0.2742531$, if z is standard normal. This means that this is not an unlikely set of events, now to consider the two sided we would double this, giving us 0.5485062, which makes what we observed or something more extreme, in either direction from the null, very likely under the standard normal. We therefore don't have evidence to reject the null. **This is not evidence in support of the null!**



P values in general

WHAT ARE THESE $\mathrm{P}(z >= s)$? they are p-values! We have all heard of p-values, but now we see they are not magic. They are simply the comparison of what you see against a known reference distribution.

You either know this reference due to some CLT or because you have made an assumption. This may not lead to a reference distribution of standard normal, regardless there are some things that are universally true about p-values.

A P-value is the probability of observing something as or more extreme than our test statistic computed from the distribution of the test statistic under the assumption that the null hypothesis is true.

WHAT THE TEST STATISTIC AND THE NULL are depends on the test, which very likely depends on the data. The meaning of the p-value however, as is stated in bold above, is always the same. The P-value is not the probability of the null. A large p-value, close to 1, does not give you evidence to support the null. Inference based on a p-value is also with regard to the DGM or super population, not the current data.

IT IS NEVER RIGHT to say that we accept the null. Or that we accepted the alternative really. We simple do or do not have evidence to reject the null, which generally means we are in the alternative.

IT IS NEVER RIGHT to say that we prove that the observed data are or are not something. We can see that the observed data are different or not. We are not testing if the sample mean of height in the males we observed is different than the sample mean of height in the females we observe. We can run a test using the data we observe, to test the hypothesis that the true DGM expectation of heights for males is different than that for females, and thus inference is for those true DGM expectations.

WHY 0.05...why not? There is no good reason. But it is accepted as something that is unlikely, and thus convincing enough to provide evidence to reject the null. But 0.05 is just a convention, there is no magic number. Is a p-value of 0.000003 more convincing than one of 0.049? Yes of course, but we would call them both significant. For this reason, there is a movement, and let us encourage you, to always report the actual p-value. report 0.049 and 0.000003, not just < 0.05.

The exception to this are p-values that are reported to you as 0. P-values can never be 0, even if your software program tells you so,

these should be reported as something like < 0.001 or even smaller depending on the number of digits you are giving in the text or table for the other p-values, or report $<$ the scientific notation for precision of your software program. Giving an example from the Z-test, a p-value of zero is not possible in truth because the standard normal has a support of $(-\infty, \infty)$, as long as your test statistics is finite then values more extreme than it have a non-zero probability of occurring if they are being generated by a standard normal random variable.

One-sided p-values are for a inequality null $\mu \leq 10$ rather than $\mu = 10$ and the alternative is $\mu > 10$, rather than $\mu \neq 10$. The boundary of this test is $\mu = 10$, the largest null μ, we use this as our point to construct a single null distribution. Now if μ is actually smaller than 10, which is possible under our null, than our Z-statistics should be even smaller than if the true μ is 10. But, if the true μ is larger the Z-statistic will be larger, therefore we want a p-value in the direction of the alternative, $\mathrm{P}(z > Z)$, where Z is a standard normal. Alternatively, if our null was $\mu \geq 10$, The boundary would still be $\mu = 10$, but this would give us an alternative of $\mu < 10$, and the p-value we want to report to reject or not reject this null would be in the direction $\mathrm{P}(z < Z)$, where Z is a standard normal. [46]. In the two-sided case, as we will see in the lab, often we will disregard the sign of the z, and simply calculate the $\mathrm{P}(|z| > Z) * 2$, where $|z|$ is the absolute value or positive amount of z. This is because, as we have discussed, the standard normal and the t are symmetric about 0, and therefore when you are calculating a two-sided p-value the sign of the test statistic does not matter provided you are comparing the correct tail.

[46] Note this is including the sign of z

Other One Sample tests

All (Frequentist) hypothesis tests take on a similar structure as the Z-test. You construct a statistic and then compare to some reference distribution to obtain inference. We will first discuss other one sample tests before moving on to the more useful two sample tests. First we will revisit the one sample Z-test to make sure that all assumptions are clear.

One Sample Z-test

The statistic is $\hat{z} = \frac{(\overline{x} - \mu_g)}{\hat{\sigma}/\sqrt{n}}$. By the CLT this is distributed standard normal under the null. The assumptions for a Z-test to give correct inference are:

1. independent and identically distributed x
   this means that x_1 and x_2 are not clearly correlated, for example they are not repeated measurements on the same person, and that the same $f(x)$ that produced x_1, produced x_2, we didn't switch DGM in the middle of data collection. This is often written as (IID)

2. large enough sample for the CLT to apply or normal data and assumed or known SD under the null.

A Z-test is performed by considering the assumptions, calculating a Z-statistics based on the data, and then comparing to a standard normal to obtain a p-value. The Z-test also has a natural thing to report an a confidence interval, which we will discuss in the next section.

One sample t-test

When we have a small sample we can not use the Z-test, unless we know the true DGM σ or are willing to assume what it is under the null and that the DGM is normal. Instead in small samples, if we know that the DGM is normal, or if we are willing to assume it, we can estimate σ with the sample standard deviation, and use the T-distribution as the reference instead of the standard normal.

The statistic is the same as the Z, but is often called T, $\hat{T} = \frac{(\overline{x}-\mu_g)}{\hat{\sigma}/\sqrt{n}}$. By a proof that is longer than you want us to go through in class, under the following assumptions, $\hat{T}$ is distributed T with degrees of freedom $n-1$.

1. IID x

2. $x \sim N(\mu, \sigma^2)$[47]

[47] Read below before writing on the exam or telling people what they can and cannot do with a t-test.

A t-test is a Z-test

In a large enough sample. When we are in asymtotia and the CLT has kicked in, the degrees of freedom are approaching ∞, and therefore the t distribution is converging to the standard normal. Therefore we can put these two test together, and merge the assumptions.

1. IID x

2. $x \sim N(\mu, \sigma^2)$ **OR** large enough sample for the CLT to apply

Then we can always compare to a t distribution, because if we are in a small sample and assuming a normality, then it works and if we are in a large sample and relying on the CLT, the t will be very very close to the standard normal, and we will basically be performing a Z-test, see Figure 15. [48]

[48] This is why no one uses a Z-test, a T-test and a Z-test are the same thing is large enough samples, and in small samples you have to assume normality regardless of which test...

One sample test of Proportion

A PROPORTION IS A SAMPLE MEAN so in a large enough sample size one can use a Z-test. [49] There is no reason that one cannot use the same test and test statistic for a proportion. By the CLT, the standard normal is still the correct reference distribution, there is no reason at all that we should not use this. We know that binary data is not normal, so the assumptions are as follows, and we can construct the test statistic in the exact same way!

[49] or a T-test because now we know in large enough sample sizes these are the same.

1. IID x
2. large enough sample for the CLT to apply

WE HAVE NOW DEFINED SEVERAL METHODS OF TESTS, but tests are not perfect. What are the properties of the test for which we should be concerned? AND why do these assumptions have to do with anything? Second question first, the assumptions ensured desirable properties. BUT what are those properties when it comes to a hypothesis test? The properties of interest are size or Type one error rate and Power or 1- Type two error rate.

Type I, II error and power

Type I error rate

THE TYPE ONE ERROR RATE is the rate at which the test rejects when the null is actually true. For any correctly sized test, the type one error rate should be the same as the level you are using to determine significance. The classical level for a two sided test is, as we know, 0.05. So a correctly sized test will reject, when in truth the null is true, 5% of the time. If your level, α, is 0.05. Then, if your test statistics is in the tail of your reference distribution, you will reject the null. However, you MAY be wrong, and the probability that you are wrong should be 0.05, because that is the amount of the null distribution that is contained in the tails. If the assumptions outlined above for each test actually hold it will ensure that for those tests The type one error rate will equal the level α.

THIS IS NOT TRUE if the assumptions do not hold or for other tests. Many many tests are wrong, reject when the null is true, more often than α. This is one of the fundamental properties of any test, if it does not have a type one error rate that is equal to α, under the stated assumptions, it is not a good test and you should not use it. This also means that you should consider the assumptions and if they are likely to hold in your setting before using a test, because it is only correctly sized under those assumptions. In formula the Type one error rate is P(type I error) =P(reject H0 | H0 is TRUE) = P(wrongly reject H0). A type I error is also referred to as 'false positive' and if P(type I error) = α, then is a correctly sized test.

WHY DO WE CARE? Examples of type one errors are rejecting the null that treatment does nothing or kills people when in fact the treatment does not kill people. Type one errors can be extremely serious and unless you care nothing for science or truth you should care about them.

AS ONE CAN see when you have a correctly sized test α is type one error rate, but that is one correctly run test under the assumptions. Now if instead you run several tests and take the smallest p-value, the global type one error rate, rejecting at least one test when the null holds for all tests, is roughly the number of tests times 0.05, running 10 tests gets you to about 0.5, meaning the rate of errors under the global null is 50%. This is why we care about pre-specifying a hypothesis and testing exactly one hypothesis unless we adjust the p-values or α level to account for the number of tests run. When you do hundreds of even thousands of tests at the 0.05 level, in most cases, the probability that one of the tests will reject is one you are no longer testing anything, rejection of at least one null is assured.

Type II error rate and Power

WHEN YOU DO NOT GAIN EVIDENCE AGAINST THE NULL HYPOTHESIS that does not mean the null is true. For example, the mean was not 0, but your test did not provide a sufficiently strong evidence in this direction, because the Z you calculated was a relatively common value obtained from a standard normal. This is a 'type II error'.

THE RISK OF TYPE II ERROR depends on: the sample size (n); and how wrong the null hypothesis is. For example, suppose you have a coin, and you want to test the null hypothesis that $\mathrm{P}(x = 1) = 0.5$. If the true value is $\mathrm{P}(x = 1) = 0.9$, it will be easy, meaning in small

samples, to detect this, i.e., obtain evidence that H0 is false. Even with a small sample, say n = 10, you will probably see that something is wrong, because x = 1 will be MUCH more frequent than x = 0. If the true value is $P(x = 1) = 0.51$, the null hypothesis, $P(x = 1) = 0.5$, is false, but it will be very difficult to detect this, i.e., you will need a very large n. When H0 is very false you will make less type II errors because you will detect this deviation from the null very easily, i.e. in small samples. More generally, a small sample will be sufficient to detect a big difference between the null and the truth, which could also be thought of as a large effect or a strong association. If the effect is small, or the association is weak, then a larger sample size is needed.

FORMALLY P(type II error) = β = P(don't reject H0 | H0 is FALSE) = P(wrongly fail to reject H0) A type II error is also referred to as 'false negative'. Power is 1-β, or rejecting when you should, or rejecting when the null hypothesis is false.

POWER is the probability of rejecting H0 when it is false, i.e., the chance of avoiding type II error, is called Power = 1 – P(type II error) = $1 - \beta$ = P(I reject H0 | H0 is FALSE) = P(correctly reject H0). It is important that you have a sufficient power: otherwise, you are wasting time and money. To achieve a sufficient power, you must select a sufficiently large value of n (sample size calculation), given a guess for the size of the difference from the null.
For example, going back to the wine seller experiment. BEFORE you go, ask yourself: 'if there is any effect, how much will it be?', OR, 'What is the smallest effect I care about?' Say that you care if there is a 1% or more difference. This is a small number: if you only have 10 bottles, you will have very low power to detect it. So, how many bottles will you need, in order to detect a 1% difference, with a power of 90%? You can do a power calculation, and find that you would need an n = 1348. This is how many bottles you should collect. If you don't have the resources, collecting less is not an option if what you really care about is a 1% difference or more. You could say that although you care about a 1% difference, given your resources you are willing to accept that you only have power to detect a 50% difference. Instead save your money and run a better experiment when you can! A general formula to compute the value of n or power is not provided here, because the procedure to apply depends on the type of test. [50]

[50] In lab you will calculate power via simulation.

Summary

P(type I error), is α the significance level for a correctly sized test. If a test does not have type I error rate of α it is a bad test. P(type II

	REJECT	DON'T REJECT
H0 TRUE	Type I error (false positive)	Great!
H0 FALSE	Great!	Type II error (false negative)

Table 3: Testing Errors type I and type II errors

error) $= \beta$ and power $= 1 - \beta$. Power and type II error rate depends on the sample size and the true effect size. One should consider the assumptions of a test, which ensure that it was correct type I error, as well as the sample size and effect size of interest, which determines the power, before running a test.

Confidence intervals for μ

I'M INVESTIGATING THE MEAN OF THE DISTRIBUTION THAT GENERATES WEIGHTS IN SWEDEN, and I collect a sample of n = 100 observations. The sample mean is $\overline{x} = 72.3$Kg, with 95% CI $(70.1, 74.5)$. This means that 'I don't know what the true mean μ is, but I 'believe' it is likely between 70.1 and 74.5.
More precisely: If I repeat the same experiment many times, and each time I compute a 95% [51] confidence interval, then 95% of the confidence intervals will cover the true mean. Of course, the true mean IS either INSIDE or OUTSIDE your specific confidence interval, and you will never know it.

[51] Question: why compute a 95% or 99% CI, and not 100% CI? Wouldn't it be nice to be 100% sure? The answer is: a 100% confidence interval would be $(-\infty, \infty)$ because the normal or T value you would use (like the 95 1.96) would be ∞. Intuitively, this CI would include extreme events which are very unlikely to occur, like 'a dice rolls 6 for 100 times in a row'.

Providing a confidence interval is useful in two ways:

1. You show that, of course, you don't believe that the mean weight in the population is exactly 72.3 Kg, and you clarify that 72.3 is just an estimate.

2. You show how precise you are: if another researcher does the same experiment, but with a smaller sample (say, n = 20), her confidence interval will be larger. For example, she will find $\overline{x} = 71.1$Kg, with 95% CI (61.5, 79.6).

3. it gives you an idea about the likely values of μ

HOW TO OBTAIN A 95% CI FOR μ, let us consider the Z test.

- We know that the z statistic is N(0,1) by the CLT, in large enough samples

- If $z \sim$ N(0,1), then P(-1.96 < z < 1.96) = 0.95

WE KNOW A WE HAVE A STANDARD NORMAL VARIABLE AND WE KNOW A PROPERTIES OF A STANDARD NORMAL. Combining this, we find:

$$\mathrm{P}(-1.96 < z < 1.96) = 0.95 = \mathrm{P}(-1.96 < \frac{(\overline{x} - \mu)}{\hat{\sigma}/\sqrt{n}} < 1.96)$$

with some algebra, we obtain:

$$\mathrm{P}(x - 1.96 * \hat{\sigma}/\sqrt{n} < \mu < x + 1.96 * \hat{\sigma}/\sqrt{n}) = 0.95$$

which is a 95% CI for μ. Informally, you can say: $x \pm 1.96 * \hat{\sigma}/\sqrt{n}$

HERE ARE THREE INTERPRETATIONS OF A SINGLE CONFIDENCE INTERVAL that are ok to use in order from good to ok [52]

[52] be careful!

- The sample mean we observe would be typical of what we expect if the true DGM μ were in the interval
- My estimate is precise or not precise based on the length of the CI. (No mention of the true μ here, good!)
- I am 95% confident that the true μ is within my interval.[53]

[53] (confusing because people will take this to mean that the true mu is in there or that the probability that it is in there is 0.95, which IS NOT TRUE). However, provided that what you mean by confident is that over repeated samples 95% of these same intervals will contain the DGM μ, then this is ok. You can say this, just don't confuse yourself!

IF THEY ARE SO HARD TO INTERPRET CORRECTLY, WHY DO WE HAVE TO COMPUTE A CONFIDENCE INTERVAL? To show that a sample quantity is an estimate, and to describe the associated variability. For example, if researcher A conducts an experiment with n = 10, and finds $\overline{x} = 135$, and researcher B replicates the same experiment with n = 1000, and finds $\overline{x} = 148$, which value is more reliable? Obviously the second one, because n was much larger. This will be reflected by the fact that A's confidence interval will be much wider than B's.

EXAMPLE OF CALCULATING A 95% CI: Suppose you have your sample of n = 100 people with sample mean weight $\overline{x} = 72.3$Kg. Say that $\hat{\sigma}_x = 11.2$.
Just apply the formula:
$\overline{x} - 1.96 * \sigma_x/\sqrt{n} = 72.3 - 1.96 * 1.12 = 70.1$
(lower bound of a 95% CI)
$\overline{x} + 1.96 * \sigma_x/\sqrt{n} = 72.3 + 1.96 * 1.12 = 74.5$
(upper bound of a 95% CI)
These are the extremes of your 95% confidence interval: (70.1, 74.5).

If you want a 99% confidence interval, replace 1.96 with 2.58. This is because, if $z \sim$ N(0,1), then $\mathrm{P}(-2.58 < z < 2.58) = 99\%$. Of course, a 99% CI will be larger than a 95% CI!!

$\overline{x} - 2.58 * \sigma_x / \sqrt{n} == 72.3 - 2.58 * 1.12 = 69.4$ (lower bound)
$\overline{x} - 2.58 * \sigma_x / \sqrt{n} == 72.3 + 2.58 * 1.12 = 75.2$ (upper bound)
99% CI is: (69.4, 75.2).

If you want a shorter confidence interval i.e. more precision, the best thing to do is to increase n. For example, if the same experiment was conducted on 900 subjects, the 95% CI would be:
$\overline{x} - 1.96 * \sigma_x / \sqrt{n} = 72.3 - 1.96 * 0.3733333 = 71.6$(lower bound)
$\overline{x} + 1.96 * \sigma_x / \sqrt{n} = 72.3 + 1.96 * 0.3733333 = 73$ (upper bound)

For the same reason you compare to the t distribution when we do the t-test is a small distribution and assume that the $x \sim \mathrm{N}(\mu, \sigma^2)$.

How to obtain a 95% CI for μ, let us consider the t-test.

- We know that the t statistic is t(n-1)
- because the student's t is symmetric about 0, if we determine the t_c such that $\mathrm{P}(t < t_c) = 0.975$ we can use the same formula to get t based 95% CI

Going back to our example above, suppose you have a sample of n = 50 people with sample mean weight $\overline{x} = 72.3$Kg. Say that $\hat{\sigma}_x = 11.2$.

We just need to find the cut-point of the t(49), such that $\mathrm{P}(t < t_c) = 0.975$, using a software program or a table it is easy to see that this is 2.009575. We can now use the same formula as above:
$\overline{x} - 2.009575 * \sigma_x / \sqrt{n} == 72.3 - 2.009575 * 1.583919 = 69.117$
(lower 95% t-based bound)
$\overline{x} - 2.009575 * \sigma_x / \sqrt{n} == 72.3 + 2.009575 * 1.583919 = 75.483$
(upper 95% t-based bound)
99% CI is: (69, 75.5).

the formula to compute a confidence interval for a mean, and the method to make a one-sample z-test, are both derived from CLT. In practice, they are almost the same thing for most tests [54]:

- If a 95% confidence interval *does not* contain the value of the null hypothesis, *then* the two-sided p-value of the test will be less than 0.05 and the null hypothesis will be rejected;
- If a 95% confidence interval *does* contain the value of the null hypothesis, *then* the two-sided p-value of the test will be more than 0.05 and the null hypothesis will not be rejected.

[54] For the Fisher's exact test and some other exact tests this will not be true as the CI and the p-value are not based on the same procedure.

Two sample estimation and inference

All tests proceed as follows: Formulate a null hypothesis H0. The null hypothesis is always something like 'there is no effect', 'no difference', or 'no association'. The goal is typically to collect data and determine whether there is evidence against H0. Compute a 'test statistic'. For example, in the one-sample t-test described above, you have the test statistic z. Use some theorem (in most cases, the Central Limit Theorem or a variation of it) to have a null distribution to compare to in order to compute p-values. If p is small, say, less than $\alpha = 0.05$, reject the null hypothesis; otherwise, don't reject.

Each test is used in a specific situation. We will discuss in detail each of these tests in the same way we did for the the one-sample t-test. For all one sample tests we are comparing one mean with a known or guessed value, e.g., to see if the mean content of wine bottles is really 1 liter as the seller claims. In other situations, the known value may come from a doctor's recommendation (e.g., compare the systolic blood pressure of my patients with the ideal value of 130), or may be derived from a previous study. In reality, the one-sample t-test is not used very frequently. Typically, we want to compare *two or more* samples, or measure associations between numerical variables.

Two sample tests for continuous data

Parametric tests for differences in the DGM expectation

The term parametric refers to the fact that there is a particular parameter of interest, the expectation. This is often confused with the test assuming a particular distribution, but one can assume a particular distribution and still run a non-parametric test...it is about if there is a parameter, like the expectation, you are attempting to do inference for. There are two common tests here.

- To compare two means, use two-sample t-test (which again becomes a two sample Z-test is large samples!)
- To compare more than two means, use ANOVA.

ANOVA is a old term and method and although I mention it here for completeness, we will not cover it is class nor will you ever need to use in reality. It someone asks for ANOVA, just run linear regression and give them a better answer.

TWO SAMPLE T-TEST assume we have two random variables x_1 and x_2 from which we have samples. The test statistic to compare there expectations for the null H0: $\mu_1 = \mu_2$ is given by:

$$t = \frac{\overline{x}_1 - \overline{x}_2}{\hat{se}}$$

the correct $\hat{se}$ and reference distribution will dependent on the assumptions you are willing to make

UNDER THESE ASSUMPTIONS:

1. The data are IID within group and the groups are independent
2. Large enough samples from both groups such that the CLT applies, or both $x_1 \sim \mathrm{N}(\mu_1, \sigma_1^2)$ and $x_2 \sim \mathrm{N}(\mu_2, \sigma_2^2)$
3. $\sigma_2^2 = \sigma_1^2$

The standard error estimate should be $\hat{se} = \hat{S}_p\sqrt{1/n_1 + 1/n_2}$, where

$$\hat{S}_p = \frac{(n_1 - 1)\hat{\sigma}_1^2 + (n_2 - 1)\hat{\sigma}_2^2}{(n_1 - 1) + (n_2 - 1)},$$

where $\hat{\sigma}_1^2$ is the sample variance estimate within group one. and the degrees of freedom for the t to compare to is df= $n_1 + n_2 - 2$. This is the pooled variance and requires the assumptions of homoskedasticity, or equal DGM variance (assumption 3).

UNDER JUST THE FIRST TWO ASSUMPTIONS:

1. The data are IID within group and the groups are independent
2. Large enough samples from both groups such that the CLT applies, or both $x_1 \sim \mathrm{N}(\mu_1, \sigma_1^2)$ and $x_2 \sim \mathrm{N}(\mu_2, \sigma_2^2)$

The standard error estimate should be $\hat{se} = \sqrt{\hat{\sigma}_1^2/n_1 + \hat{\sigma}_2^2/n_2}$. This makes the degrees of freedom complicated, one calculation of the degrees of freedom is given by:

$$\mathrm{df} = \frac{(\hat{\sigma}_1^2/n_1 + \hat{\sigma}_2^2/n_2)^2}{\frac{(\hat{\sigma}_1^2/n_1)^2}{(n_1-1)} + \frac{(\hat{\sigma}_2^2/n_2)^2}{(n_2-1)}}.$$

So, by reducing the number of assumptions we make the $\hat{se}$ less complicated and the degrees of freedom calculation much much more complicated. However, this test is robust against the third assumption above, meaning that assumption is not needed, and this is being robust against heteroskedasticity, which is the opposite of homoskedasticity.

YOU COULD CHECK IF, within each group, the data are Normal. You could make a histogram or a density plot, or use Shapiro-Wilk test for Normality. The null hypothesis of Shapiro-Wilk test is normality, and in this case you are happy if you don't reject it. In reality, it is not so important that the data are truly Normal (which will never happen in practice): it will be ok if the data are reasonably symmetric, and with no outliers (extreme values).

In practice if the data are small enough that you need to actually worry about normality, testing for normality makes very little sense because you will generally not have the power to reject. But, if a reviewer asks you, you can test.

YOU MAY USE BARTLETT'S TEST, for example, to see if the variances are significantly different between the two groups. Hopefully they are not. The null hypothesis of Bartlett's test is homoskedasticity, and in this case you are happy if you don't reject it. If n is very large (e.g., n = 1000). Instead just use the robust t-test and don't make this assumption!

Two sample non-parametric tests

THE IDEA IS TO COMPARE not just the mean of two or more groups, but the entire distribution. It is like to compare two or more histograms. This can be done by the following tests:

- Compare two distributions: Mann-Whitney-Wilcoxon test, aka the Mann-Whitney-U test, Wilcoxon rank sum test. This is the nonparametric equivalent to two-sample t-test
- Compare more than two distributions: Kruskal-Wallis test. This is the nonparametric equivalent to ANOVA

In the two sample case, consider again that there are two random variables x_1 and x_2 from which we have samples. The null hypothesis for the Mann-Whitney-U test is that x_1 and x_2 have the same distribution. In other words, if we randomly select two samples, one from x_1 and one from x_2, then it should be equally likely that the

first sample is numerically smaller than the second. So another way to state the null hypothesis is that $Pr(x_1 < x_2) = 0.5$ (or equivalently $Pr(x_1 > x_2)$). We can compute the corresponding sample proportion of that probability by enumerating all possible pairs and counting the number where the sample from x_1 is less than that from x_2:

$$U = \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} I[X_{1i} < X_{2j}].$$

Under the null, this proportion has the *Wilcoxon* distribution. In large samples, it is approximately normally distributed. The calculation can be done more efficiently using ranks of the data.

MANN-WHITNEY-U test is run as follows:

- Assign numeric ranks to all the observations (put the observations from both groups to one set), beginning with 1 for the smallest value. Where there are groups of tied values, assign a rank equal to the midpoint of unadjusted rankings. E.g., the ranks of (3, 5, 5, 5, 5, 8) are (1, 3.5, 3.5, 3.5, 3.5, 6) (the unadjusted rank would be (1, 2, 3, 4, 5, 6)).

- Now, add up the ranks for the observations which came from sample 1. The sum of ranks in sample 2 is now determinate, since the sum of all the ranks equals N(N + 1)/2 where N is the total number of observations.

- U is then given by:

$$U_1 = R_1 - \frac{n_1(n_1+1)}{2}$$

  where n1 is the sample size for sample 1, and R1 is the sum of the ranks in sample 1.

- Note that it doesn't matter which of the two samples is considered sample 1. An equally valid formula for U is

$$U_2 = R_2 - \frac{n_2(n_2+1)}{2}$$

  Assumptions:

1. All the observations from both groups are independent of each other,

2. The responses are ordinal (i.e., one can at least say, of any two observations, which is the greater),

The test involves the calculation of a statistic, usually called U, whose distribution under the null hypothesis is known. In the case of small samples, the distribution is tabulated, but for sample sizes above ~ 20, approximation using the normal distribution is fairly good. Some books tabulate statistics equivalent to U, such as the sum of ranks in one of the samples, rather than U itself.

These tests are said to be 'nonparametric', because they use a weaker version of CLT and require less assumptions. In particular, they are based on the rank, i.e., they don't use the original data but only their position in the sample. You don't need to worry about and homoskedasticity or normality, but unless you use exact p-value you still need a large enough sample size. The disadvantage of these tests is that, in general, they are less powerful than the parametric tests when the parametric tests could have been used (i.e., they have a lower probability of correctly rejecting the null hypothesis when it is false).

Frequently, you will draw similar conclusions from a parametric and a nonparametric test (e.g., the p-value of a t-test will be close to that of a Wilcoxon test). However, this is not always the case because they are testing different hypotheses. If two samples have the same mean, but different variance, then the null hypothesis of the t-test is true, but the null hypothesis for the Wilcoxon test is false!

Nonparametric tests are preferable if the sample size is small. However, if n is very very small, e.g., $n = 5$, then you have no chance to detect a difference with a nonparametric test and you may decide to go for a parametric one. In this case you must simply assume you have data from a normal, run the test and justify your assumption. You can't test here, you don't have enough data. The best thing to do is avoid this situation!

A remark around measures of change. Assume you want to compare two groups, in which group 1 and group 2 are actually composed of the SAME patients: group 1 is 'before treatment' and group 2 is 'after treatment'. In this case, the data are not independent, and to compare the two means you should use **paired t-test**. If you have 3 or more repeated observations, e.g., the same patients are observed at times $1, 2, 3 \ldots$ then you can compare means by **repeated-measures ANOVA** (parametric) or **Friedman's test** (nonparametric), but there are a lot of other things you can do in this situation as well. This is beyond the scope of this class, but if you have such data consider using generalized estimating equations.

	Exposed	Unexposed
Cases	10	90
person years	100	900

Table 4: Example two-by-two table

test for differences in rates

or association between a binary variable and a count. Above we considered incidence rates and differences in rates and rate ratios.

SUPPOSE YOU HAVE THE FOLLOWING DATA:

The estimated incidence rate per person year in the exposed is $10/100 = 0.1$, we may wish to put a confidence interval on this. We should log it, log(incidence rate) = -2.302585. The the se(log(incidence rate)) $= 1/\sqrt{10}$ or in general 1 over the square root of the number of events. we can then construct an 95% CI assuming large enough sample size via the standard normal approximation on the log scale as $-2.30 \pm 1.96 * 0.316$, and then take e raised to those values to put the CI back on the orginal scale. This makes the 95% CI for the incidence rate of 0.1, (0.0540, 0.186).

TO TEST FOR AN ASSOCIATION BETWEEN EXPOSURE AND INCIDENCE or to test for a difference in incidence rates between the unexposed and exposed. [55] we can either test for a difference in rates, or a rate ratio.

[55] this is are same question when using this test, they are just stated differently

To test for the null that the difference in rates is 0, i.e. difference is rates = Rate per unit in group a − Rate per unit in group b = 0, can be run using a t-test under the assumptions:

1. Large enough sample in both groups
2. iid data within group and independence between groups

to test for the null that ratio of rates is equal to 1, i.e.

$$\text{Rate per unit in group a/Rate per unit in group b} = 1$$

we could again use the t-test on the log of this statistic, because the log(a/b)=log(a)-log(b), however, under the assumption that the rates are Poisson there is a better estimate of the standard error than the one used in the t-test, although in large enough samples it will not matter. This is:

$$se(log(RR)) = \sqrt{\frac{1}{10} + \frac{1}{90}}$$

Resulting in a z-statistic of:

$$\frac{log(RR)}{se(log(RR))}$$

which under the assumptions of:

1. Large enough sample in both groups
2. IID data within group and independence between groups

we can compare this to a standard normal to obtain a p-value. Similarly, we can use the standard normal approximation to obtain 95% CI in the same way as we did in the one sample case. In this case the log(10/100)-log(90/900) is zero, and the rate ratio is 1. There is no need to do a test, the p-value is 1. However, we can get a confidence interval to see how precise this estimate is, $0 \pm 1.96 * 0.33$, which is (-0.65, 0.65) after raising e to these values, $(e^{-0.65}, e^{0.65})$ we find that the 95% CI for the rate ratio is (0.52, 1.92).

Tests of correlation

When you have two numeric variables and you wish to know about their association, consider the correlation.There is the linear correlation, which is called the Pearson correlation coefficient.

$$\hat{r}_{xq} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(q_i - \bar{q})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(q_i - \bar{q})^2}}$$

Assumptions:

- x,q are bivariate normal or large enough sample size
- linear relationship between x and q
- the (x_i, q_i) pairs are iid

If you assume x,q are bivariate normal, then one can construct a t distributed test statistics.

$$t = \hat{r}_{xq}\sqrt{\frac{(n-2)}{1-\hat{r}_{xq}^2}},$$

which has a t distribution with $n-2$ degress of freedom.

When the sample size is sufficiently large, then a Z-test statistic can be constructed. Using the Fisher's transformation,

$$\mathrm{F}(r) = \frac{1}{2}\ln\frac{1+r}{1-r}$$

The Z-test statistic for the null $\rho = \rho_0$ is given by[56]:

$$[\mathrm{F}(\rho) - \mathrm{F}(\rho_0)]\sqrt{(n-3)}$$

[56] There are multiple ways to calculate the exact or approximate reference distribution.

IF YOU DON'T BELIEVE THERE IS A LINEAR RELATIONSHIP between the two numeric variables, then consider using the rank-based correlation as a summary of the association. We first need to rank the variables, then calculate the Pearson correlation of the ranks. This is the Spearman's rank correlation coefficient.

$R_x = \text{Rank}(x)$ $R_q = \text{Rank}(q)$

$$\hat{r}_{R_x R_q} = \frac{\sum_{i=1}^{n}(R_{xi} - \bar{R_x})(R_{qi} - \bar{R_q})}{\sqrt{\sum_{i=1}^{n}(R_{xi} - \bar{R_x})^2}\sqrt{\sum_{i=1}^{n}(R_{qi} - \bar{R_q})^2}}$$

Assumptions:

- Large enough sample size
- the (x_i, q_i) pairs are iid

NOTE THAT REGARDLESS OF THE LINEAR OR RANK BASED CORRELATION ρ_{xq} is always between -1 (perfect negative correlation) and +1 (perfect positive correlation), and is zero if there is no correlation (except in some particular conditions, in which there is association between x and q, but the correlation is zero.

A CORRELATION TEST WILL TEST THE FOLLOWING NULL HYPOTHESIS: H0: $\rho_{xq} = 0$, which usually is interpreted as x and y are independent (but pay attention to situations like that in the above figure, where cor = 0 but the two variables are NOT independent). As usual, you will get a p-value: if it is small, you will reject H0 and conclude that there is a significant association.

THIS IS A PRETTY SILLY TEST almost any two things measured on the same groups of people in a large enough sample will be significantly correlated.

Tests for two binary variables

Consider you have two binary variables with cell counts as below, a, b, c and d.

	Group 1	Group 2	Total
Number with positive outcome	a	c	a+c
Number with negative outcome	b	d	b+d
Total	a+b	c+d	a+b+c+d

ASSUMPTIONS FOR ALL TESTS BELOW:

1. both binary variables are iid and the two groups are independent
2. large enough sample size for the CLT

tests for difference in proportion, risk difference

In a large enough sample size you can simply use a Z-test or t-test. This is formulated in the same way as the two-sample t or z test.

Formula for risk difference using the above two-by-two table is given by:

$$\text{RD} = \frac{a}{(a+b)} - \frac{c}{(d+c)}$$

Allowing for unequal variance, in a large enough sample size one can use the same se estimate as in the t-test for two samples. In large sample and a large enough number of events, $n * p > 5$, you can use the se formula the risk difference. First calculate the pooled proportion, $\hat{p}$, $(a+c)/(a+b+c+d)$, then the standard error

$$\text{se(RD)} = \sqrt{\hat{p} * (1-\hat{p}) \frac{1}{(a+b)} \frac{1}{(b+d)}}$$

This can be compared to a standard normal distribution, under the additional assumption of the same variance, which would be the case if the expectations where the same!

tests for difference in log proportions, relative risk

Formula for relative risk using the above two-by-two table is given by:

$$\text{RR} = \frac{a/(a+b)}{c/(d+c)}$$

We need to natural log transform the RR, ln(RR) we then have a formula for the standard error, $\hat{se}(\ln(\text{RR}))$ is given by

$$\sqrt{\frac{1}{c} + \frac{1}{a} + \frac{1}{(a+b)} + \frac{1}{(d+c)}}$$

Then you can compute a Z-statistic and compare to the standard normal.

$$z = \frac{\ln(\text{RR})}{\sqrt{\frac{1}{c} + \frac{1}{a} + \frac{1}{(a+b)} + \frac{1}{(d+c)}}}$$

tests for difference in log odds, odds ratio

Formula for odds ratios using the above two-by-two table is given by:

$$\text{OR} = \frac{a/b}{c/d}$$

e need to natural log transform the RR, ln(OR) we then have a formula for the standard error, $\hat{se}(\ln(\text{OR}))$ is given by

$$\sqrt{\frac{1}{c}+\frac{1}{a}+\frac{1}{b}+\frac{1}{c}}$$

[57]

Then you can compute a Z-statistic and compare to the standard normal.

$$z = \frac{\ln(\text{OR})}{\sqrt{\frac{1}{c}+\frac{1}{a}+\frac{1}{b}+\frac{1}{d}}}$$

[57] OR must be used in case-control studies as they are invariant to the selection of which is the outcome and which is the exposure. i.e. you get the same OR if you calculate the odds of y given x as you do if you calculate the odds of x given y.

Chi-square test

A test for association the chi-square test compares the observed frequencies to what would be expected if the two binary variables were independent.
from the two-by-two table above the expected cell counts numbering the cells from upper left to lower right, $\{1, 2, 3, 4\}$ would be

$$\text{E}_1 = (a+b)*(a+c)/(a+b+c+d)$$

$$\text{E}_2 = (a+b)*(b+d)/(a+b+c+d)$$

$$\text{E}_3 = (c+d)*(a+c)/(a+b+c+d)$$

$$\text{E}_4 = (c+d)*(b+d)/(a+b+c+d)$$

$$\chi^2 = \sum_j \frac{(\text{O}_j - \text{E}_j)^2}{\text{E}_j}$$

Assumptions:

1. both binary variables are iid and the two groups are independent
2. large enough sample size for the CLT

The reference distribution is χ^2 with degrees of freedom 1, which is number of rows less 1 times the number of columns less 1, so for a two-by-two table is 1. From this, just like comparing to the standard normal we can calculate a p-value and for a set level and reject the null if the observed differs from the expected.

This can be easily expanded to one binary and one categorical variable. There are LOTS of chi-square tests, this is just one of them.

The two sample test of almost anything

A PERMUTATION TEST uses a computational approach to constructing a null distribution. The basic steps are as follows:

1. Calculate a test statistic of interest
2. Break the relationship of interest in the data, and re-sample to create a data set under the null
3. Re-calculate the test statistic
4. Repeat until you have all possible combinations or a very large number of combinations to create a null distribution

THE SAME IDEA AS ALL OF THE TESTS ABOVE, We have a null hypothesis to test we obtain a test statistic. We need a null distribution of the test statistic to compare to, but we don't know what the null distribution is! If we can enforce the null on our sample, we can simulate under the null and create a null distribution to compare to.

EXAMPLE WITH THE MEDIAN: I have a sample from a random variable x, of average heart rate for 60 seconds in two groups. One group has had an intervention of regular exercise for the last year and a different group doing whatever they normally do.

The null hypothesis is that there is no difference between the groups in median heart rate. We only know that one CLT (at this moment) and it doesn't seem to apply here because it talks about the mean, not the median...so what do we compare the median to? What is its standard error? [58] What is our test statistic?

[58] there is totally a CLT that applies, but just go with us!

HERE IS SOMETHING TO TRY Use the median as your test statistic enforce the null, no difference between the groups, by switching the labels on people, randomly, and recalculating the median do this so that every possible way this could be done, is done, including the way you observed in your study.

For large samples doing it a large number of times, but not in every possible way, is probably OK. Compare what you found to that distribution, calculating a p-value by the proportion of things that are more extreme (2-sided) or larger (greater than 1-sided) or smaller (less than 1-sided). [59]

[59] Note you include the observed value in the distribution, so it should NEVER be 0.

EXAMPLE 1:

sample 1 $= \{1, 2\}$

sample 2 = $\{6, 7\}$
Median sample 1 = 1.5
Median sample 2 = 6.5
The difference in medians = -5

Under the null it doesn't matter what sample you are in, so let's just ignore the labels and see what we get. the full sample is 1,2,6,7, we can reorder this in 5 news ways that will give us new labels for at least some of the subjects
sample 1 new = $\{1, 6\}, \{1, 7\}, \{6, 7\}, \{2, 6\}, \{2, 7\}$,
sample 2 new = $\{2, 7\}, \{2, 6\}, \{1, 2\}, \{1, 7\}, \{1, 6\}$.,

the null difference in medians is then,
$\{-1, 0, 5, 0, 1, -5\}$ This include the -5 that we observed, we compare, for a one-sided p-value we have 1 value that is less than or equal to our observation $1/4 \approx 0.17$. for a two sided p-value we have 2 values that are more extreme or equal for in this case $2/6 \approx 0.33$.

This is an easy example because we can do all the cases, easily! but when you have a lot more data, what do you do? you can try to do all cases, or you can just can a large random sample without replacement, or switch the labels lots of times, keeping the number in each group constant.

Example 2:
Sample 1 is a random sample of 100 from a N(1,1)
Sample 2 is a random sample of 100 from a exp(1)
we want to test for a difference in medians. We want to permute them are A LOT, 19,900 ways to put 200 people into 2 groups when order does not matter.

From our sample the median difference is 0.45. Using 100 random switches for our particular example we get a p-value 0.02, with 1000 we get 0.004, with 10000 0.002 this is nearly the same for 20,000 and 100,000. How many do you need to run? From the 20,000 random switching run we get the histogram shown to the right.

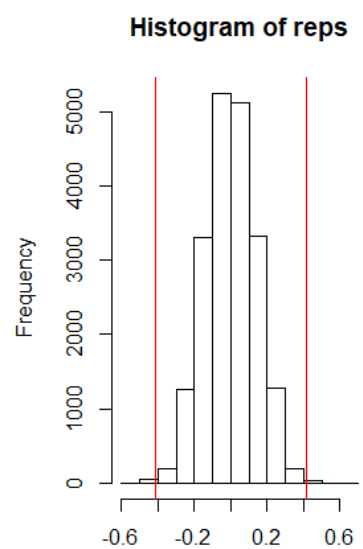


Figure 21: The permutation distribution of the null

Table 5: Summary of Two Sample Tests

Name	H0	Interpretation	Type	Assumptions for inference
Two-sample t-test	$\mu_1 = \mu_2$	Compare means of two groups	Parametric	n is large enough or data are Normal, data are IID within group and the groups are independent
Mann-Whitney-Wilcoxon rank-sum test	Two groups have the same distribution	Compare distributions of two groups	Non-parametric	data are IID within group and the groups are independent and ordinal data
Kruskal-Wallis test	k groups have the same distribution	Compare distributions of k groups	Non-parametric	data are IID within group and the groups are independent and ordinal data
Relative rate or rate ratio RR	groups have the same rate	comparing rates between two groups	Parametric	data are IID within groups and are independent between groups, large enough sample.
Pearson's linear correlation test	$\rho_{xy} = 0$	Test association between two numerical variables	Parametric	x and y have a linear association and data are IID within group
Spearman's rank correlation test	$\rho_{xy} = 0$	Test association between two numerical variables	Non-parametric	data are IID within group
Risk difference, Risk Ratio test	$p_1 = p_2$ or $p_1/p_2 = 1$	the probability of success in the two groups are equal	parametric	data are iid within groups and large enough sample size.
OR test	$odds_1 = odds_2$ or $OR = 1$	the odds of success in the two groups are equal	parametric	data are iid within groups and large enough sample size.
Chi-square test	independence	the two binary variables are independent	Non-parametric	data are iid within groups and large enough sample size.
Permutation test	Many things!	Many Things!	Parametric, but distribution free	you can simulate the null

Causal inference, confounding, mediators and effect modification

We have considered hypotheses about the DGM without formally stating a causal relationship. However, most hypotheses are about causation. In some cases you only wish to determine if there is an association or correlation between the DGM and not if this is due to a particular cause. However, often even when this is the hypothesis stated, in reality what people actually want to do is make causal statements.

It is difficult to discuss causality without additional notation, but we will attempt to do so with directed acyclic graphs or DAGS. In a DAG, the random variables are nodes and there are arrows that go from a parent variable to a child variable to indicate a causal effect. A DAG is an assumed relationship between observed and sometimes unobserved variables, which in some cases we can test. A DAG allows us to easily depict both the relationships we want to test and the assumptions we are making or need to make about this relationship to test it.

We have all heard of confounding, and we have all seen people run an adjusted analysis to account for confounding, but then still discuss the association between two variables. Unless your hypothesis is concerned with causality, you do not need to be concerned with confounding. Confounding only exists for a causal relationship, statistical associations cannot be confounded. Statistical associations can be caused by confounding, i.e. common causes of the two variable in question, but if you are interested in an association, that association cannot be confounded. One can adjust for other variables, and report the adjusted association, but what was the point of that adjustment unless you were attempting to control confounding and therefore obtain a causal effect?[60]

[60] There are good answers to this, but confounding is not one of them.

DAGS ARE ALSO A WAY TO ILLUSTRATE your sampling or study design. If without randomization of an exposure there would be a common cause between it and the outcome, but you have randomized, you would not include that common cause in the DAG for your setting. In your setting, the exposure and outcome do not have a common cause, a confounder, if you have randomized properly. If the DAG in the first figure below depicted the true relationship of the exposure x and the outcome y in the population, the next figure would depict the DAG for a randomized clinical trial for the same exposure and outcome. Randomization removes the arrow from c to x.

$$\begin{tikzcd} & c \arrow[dl] \arrow[d] \\ x \arrow[r] & y \end{tikzcd}$$

Figure 22: There is a causal effect of x on y, but there is confounding of the relationship between x and y by c.

This might be a cohort study where you know older people take more medication and are generally more sick and likely to die. If you are interested in the relationship between taking more than 6 medications, x, and death, y, then being older, c, is a confounder of that relationship.

$$x \longrightarrow y$$

Figure 23: There is a causal effect of x on y, there is no confounding of the relationship between x and y.

A perfectly randomized experiment would look like this, although there would likely be other things that impact y in addtion to x.

$$x \longrightarrow m \longrightarrow y$$

Figure 24: There is a causal effect of x on y, there is no confounding of the relationship between x and y, but there is a mediator m between x and y

This might be a randomized clinical trial where the treatment, x, causes headaches, m, and headaches impact blood pressure, y. X still impacts blood pressure, but this is mediated via headaches.

$$\begin{tikzcd} & q \arrow[d] \\ x \arrow[r] & y \end{tikzcd}$$

Figure 25: There is a causal effect of x on y, there is no confounding of the relationship between x and y, but there is a different variable q that also impacts y, this may be effect modification.

Unfortunately, showing effect modfication is not as easy as showing confounding on a DAG, and this is only one example of how this

could be drawn. [61] This might be a randomized experiment where the treatment works for men, but has no effect for women.

In the first DAG, c can also be an effect modifier of the relationship between x and y. That is, you can have effect modification without confounding, confounding without effect modification, or both effect modification and confounding by the same variable.

[61] please see Four Types of Effect Modification: A Classification Based on Directed Acyclic Graphs Vander-Weele, Tyler J.; Robins, James M. Epidemiology Issue: Volume 18(5), September 2007, pp 561-568 for greater detail on this point.

In the first DAG the causal relationship between x and y is clear and we could use any of the above tests, based on what we think the data are and the sample size we have from x and y. DAGs depict the underlying DGM, and this includes the study design. Outside a randomized trial, there may be many things that impact both x and y, but in a perfectly randomized trial, nothing impacts x other than randomization. y may still have several other variables that impact it, but as long as they are unrelated to x, we can ignore them. Let us consider that without intervention, DAG two is the DGM of x and y, randomization breaks the relationship between x and c. This effectively makes the true DAG into the fourth DAG. [62]

[62] If we measure it, we may still want to adjust for c as it will likely increase precision.

In the second DAG there is a measured confounder of the causal relationship between x and y. One way to deal with a measured confounder is to 'adjust' for it i.e. include this variable in the model for y. We are then breaking the accounting for the relationship between c and y, when evaluating the relationship between x and y, as we will discuss in the next Chapter.

In the third DAG m is between x and y, if we are interested in the impact of x on y, we should ignore m. If we are only interested in the impact of x on y that is not through m, then we should adjust for m in the model for y. Just as above, this effectively removes the arrow from m to y when considering the arrow from x to y, which in this DAG removes any relationship between x and y.

In the fourth DAG both q and x impact y. If you only care about the impact of x, q can be ignored, if you are interested to know if the impact of x on y changes with q, you may wish to fit an interaction model or fit models for y given x for each level of q. We will discuss this in the next Chapter in greater detail.

Linear Regression is Everything

Why regression? All methods so far use one variable, or two variables at the time. With regression we want to see the effect of many x-variables on a response variable, y, as well as single variables. The variable y is called the response, outcome, or dependent variable and x's are called predictors, covariates, or independent variables.

Things we do with a single variable:

- y numerical: compute the mean, quantiles, variance, etc. Do a histogram, a density plot, or a boxplot.
- y categorical (or binary): compute proportions, do a table. Do a barplot.

Things we do with two variables

- y numerical, x categorical: compute mean or quantiles of y, by x.
  - Compute mean or quantiles of y, by x.
  - Do a boxplot or histogram or density plot of y, by x.
  - Do a t-test or a Wilcoxon test (if x is binary).
  - Do Kruskal-Wallis test (if x has more than two categories).
- y and x are categorical: do a table and perform χ^2 (chi-squared) test.
- y and x numerical: do a scatterplot, compute Pearson's or Spearman's correlation, do a correlation test.

Things we do with many x variables

The group of x's can include numerical, binary, and categorical variables. Do a regression!

- If y is numerical continuous, and your sample is large enough, linear regression including any set of the Xs is a common form of regression

- If y is numerical discrete, and your sample is large enough, linear regression including any set of the Xs is a common form of regression.

- If y is binary, and your sample is large enough, linear regression including any set of the Xs is a NOT common, but IS a perfectly valid form of regression for looking at risk differences. [63]

- If y is categorical, you need to transform the data to binary. Then, if your sample is large enough, linear regression including any set of the Xs is a NOT common, but IS a perfectly valid form of regression for looking at risk differences.

[63] For binary and probably numeric discrete you should use robust standard errors, we will discuss that soon!

There are some places you probably shouldn't use linear regression. Time-to-event data, particularly when there is random censoring (not just at the end of follow-up). Standard linear regression can not deal with this, regardless of sample size. [64] When you are not interested in the difference in means you shouldn't use linear regression, in general, although there are ways to transform the data to get at slightly different parameters. Finally, you should not use linear regression when you know there is no chance the data came from a normal DGM and you have a small sample size, there is just no way around having to assume a DG distribution and fitting that type of model to the data in small sample sizes....what is small changes with the type of data. But large enough even for binary data is surprisingly small for univariate analysis. The problem with linear regression is that in general it is not univariate (i.e., it includes more than one predictor) and you must take into account the number of predictors when considering how large a "large enough" sample size is.

[64] There are data pre-processing steps that do make this possible, so I am not saying linear regression is useless here – OH NO, linear regression is never useless! – but, that using it directly on censored data probably isn't a great idea.

Univariate linear regression

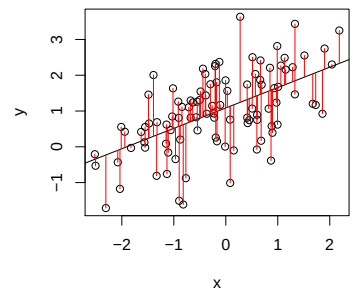
Simple Linear regression models or Univariate linear regression models contain 1 independent variable. Let us consider a model including a numeric or a binary independent variable. The mean model we are assuming is given by

$$E\{y|x\} = \beta_0 + \beta_1 x$$

This means that we assume that the DGM expectation of y is defined by the function $\beta_0 + \beta_1 x$, which is a straight line with interecept on the y axis of β_0 and slope β_1. this gives the model

$$y_i = \beta_0 + \beta_1 x + \epsilon_i$$

where ϵ_i is the residual error. Our estimates of interest are the βs. But how do we estimate them? We use ordinary least squares. Our goal is to reduce as much as possible the quantity $\sum \epsilon_i^2$, which minimizes the total distance from the line to the points.



There is a closed form for the βs with this property, which can be expressed in matrix notation.

Consider that all observed value of x are in a vector $\underline{x}$. For example if you observe x values $\{1, 2, 3, 4\}$ then

$$\underline{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix},$$

because we want to find the intercept as well, we add a vector of 1s. so that now our data matrix is given by

$$\underline{x} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}$$

then the estimate of the βs is given by

$$\widehat{\beta} = (\underline{x}^T \underline{x})^{-1} \underline{x}^T y,$$

where C^T is the transpose of matrix C, meaning the switching of row and columns and C^{-1} is the inverse of matrix C, meaning that by matrix multiplication $C^{-1}C$ is equal to the identity matrix, which is a square matrix with 1s on the diagonal and 0's everywhere else. [65]

[65] If you are lost that is fine, but you should be able to at least in theory do any type of analysis you use by hand if needed. If you don't, do you really have any idea what you are actually doing?

Example with $\underline{x}$:

$$\underline{x} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}$$

$$\underline{x}^T = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{bmatrix}$$

$$\underline{x}^T * \underline{x} = \begin{bmatrix} 4 & 10 \\ 10 & 30 \end{bmatrix}$$

How do we get this? Well, for the first cell, you take the first column of $\underline{x}^T$ and the 1st row of $\underline{x}$ you multiple the elements and then add them together. so $1 * 1 + 1 * 1 + 1 * 1 + 1 * 1 = 4$, this is is cell (1,1), then cell (2,1) is the second column of $\underline{x}^T$ and first row of $\underline{x}$

processed in the same way $1*1+2*1+3*1+4*1=10$, this is the same here for cell (1,2) as we take the 1st column and the second row and because we have $\underline{x}^T$ and $\underline{x}$ this simply $1*1+2*1+3*1+4*1=10$, then cell (2,2) is given by $1*1+2*2+3*3+4*4=30$

Now we need to take the inverse, for a 2-by-2 matrix you do this by swapping cell(1,1) and cell(2,2) and make cell(1,2) and cell(2,1) negative, then divide the whole thing by (cell(1,1)*cell(2,2) - cell(1,2)*cell(2,1)), so here,

$$(\underline{x}^T * \underline{x})^{-1} = \frac{1}{120-100}\begin{bmatrix} 30 & \text{-10} \\ \text{-10} & 4 \end{bmatrix} = \begin{bmatrix} 1.5 & \text{-1/2} \\ \text{-1/2} & 0.2 \end{bmatrix}$$

Then we can proceed with the calculations as we know how to do the rest of the steps, up to y. Let y here equal

$$y = \begin{bmatrix} 10 \\ 20 \\ 30 \\ 40 \end{bmatrix}$$

Then you can show that $(\underline{x}^T\underline{x})^{-1}\underline{x}^T y$ is $\widehat{\beta} = \begin{bmatrix} 0 \\ 10 \end{bmatrix}$, where $\widehat{\beta}_0 = 0$ and $\widehat{\beta}_1 = 10$.

GOING back to the model we know that

$$\mathrm{E}[y|x] = \beta_0 + \beta_1 x$$

.

Then $\mathrm{E}[y|x=0] = \beta_0$ in words β_0 is the expected value of y when $x=0$ and $\beta_1 = \mathrm{E}[y|x=\delta+1] - \mathrm{E}[y|x=\delta]$ is difference in the expected value of y for a 1 unit change in x. Our OLS fits $\widehat{\beta}_0$ and $\widehat{\beta}_1$ are estimates of these conditional expectations.

The matrix formulation of $\widehat{\beta}_1$ is equivalent to

$$\widehat{\beta}_1 = \frac{\mathrm{Cov}(x,y)}{\mathrm{Var}(x)}$$

.

IF X IS BINARY the meaning is the same, i.e. the expected change for a one unit change in x, but as there is only one unit of x, therefore it can also be thought of as the difference in means: $\beta_1 = \mathrm{E}[y|x=0] - \mathrm{E}[y|x=1]$.

WE HAVE NOW CONSTRUCTED estimates using OLS estimation, but how do we perform inference on these estimates and what is the null

hypothesis? Just like the Z-test or t-test, we can estimate the standard error of the coefficients and compare it to a standard normal or a t-distribution, but how do we estimate the standard error, and what assumptions allows us to use the standard normal or a t-distribution as a reference distribution using that estimate?

ASSUMPTIONS FOR CORRECT INFERENCE:

1. The pairs (y_i, x_i) are iid.
2. Exogenous error $E\{\epsilon|x\} = 0$.
3. $\text{Var}\{\epsilon|x\} = \sigma^2$. This property is called homoscedasticity. Note that this does not depend on x.
4. Large enough sample size or $\epsilon|x \sim \text{N}(0, \sigma^2\mathbb{I})$.
5. Correct specification. The linear functional form must coincide with the form of the actual data-generating process.

The variance of $\hat{\beta}$ is given by

$$\text{Var}[\widehat{\beta}|\underline{x}] = \sigma^2(\underline{x}^T\underline{x})^{-1},$$

so that

$$\widehat{se}(\widehat{\beta}_1) = \sqrt{\hat{s}^2(\underline{x}^T\underline{x})^{-1}_{11}},$$

where

$$\hat{s}^2 = \frac{\sum \epsilon^2}{n-p}$$

and p is the number of variables is included, so here, that is 1.

FOR THE NULL HYPOTHESIS THAT $\beta_1 = 0$ we can construct a test statistic

$$\frac{\widehat{\beta}_1}{\sqrt{\hat{s}^2(\underline{x}^T\underline{x})^{-1}_{11}}}.$$

In finite samples, under the assumption of normality of the residuals, this is distributed t. In large enough samples, regardless of the the underlying DGM this is distributed standard normal. Comparing to the appropriate distribution we can obtain a p-value for the null of $\beta_1 = 0$. Just like the t-test setting, if you are willing to make the needed assumptions for your sample size, a t-distribution is never wrong, which is why many stats programs use it for coefficient hypothesis testing regardless of sample size. [66]

[66] Please see figure 15

INTERPRETATION for you is much more important!

EXAMPLE 1:

We looked up and recorded 1000 days average temperatures, y, from Stockholm for random months days and years. Two student performed this random sampling student 1 and student 2 and x is 0 for student 1 and 1 for student 2. Using these data we run a linear regression for the association between y and x, hoping that the student that recorded the temperature has no impact on the DGM of the temperatures recorded. Below are the results from running the following model.

$$\text{E(Temp|Student2 indicator)} = \beta_0 + \beta_1 * \text{Student2 indicator}$$

	Estimate	Std. Error	t value	P-value
$\widehat{\beta}_0$	10.056	0.041	243.120	<0.001
$\widehat{\beta}_x$	0.009	0.059	0.158	0.875

Table 6: The example coefficient matrix from a linear regression with on binary indicator.

This means that, when x = 0, the mean of y is β_0; and when x = 1, the mean of y is $\beta_0 + \beta_1$. This also means that β_1 is the difference between the mean of y in the two groups.

The p-values are for the null hypothesis

H0: $\beta_0 = 0$

H0: $\beta_1 = 0$

Because the p-value associated with β_0 is <0.001, we reject the null hypothesis at the 0.05 level that $\beta_0 = 0$, but what does this mean really? We have evidence to support the hypothesis that the mean temperature is not zero for student 1. The p-value associated with β_1 is 0.875, which means we cannot reject the null $\beta_1 = 0$. However, we cannot say that this is evidence of the null. [67]

[67] ever, not one time, not for any reason ever.

WE MAY HAVE NOT REJECTED BECAUSE there really is no difference in the DGM expectations from student 1 and 2, or the DGM may differ so slightly that even in a sample size of 1000 we still do not have a power to detect it. If we are in the 1st case we have correctly not rejected the null, because the null is true, in the latter case we have made a type II error and with a large enough sample size we would have rejected. We don't know which of these is true when we don't reject. It is also possible that our assumptions do not hold in these data and that we do not reject when in fact we should because of a violation. [68]

[68] we can attempt to rule this out, with model checking below

Just like in a two sample t-test you are comparing the two means, and β_1 will equal the difference in means from a t-test. [69]

[69] In particular, since the linear regression model assumes equal variances, it will be equivalent to a t-test in which the variance in the two groups is assumed to be the same. This is the assumption of that $\text{Var}(\epsilon|x) = \sigma^2$, homoscedasticity. If we don't wish to make this assumption, we do not need to use a different model in general, we simply need to use a different estimate of the standard error see in section below.

OUR ESTIMATES 10.056 for β_0 and 0.009 for β_1 are the estimated expected value of temperature recorded by student 1 and the difference between expected temperatures recorded by student 1 and student 2. The estimated expected value of temperature recorded by student 2 is given by $\widehat{\beta}_0 + \widehat{\beta}_1$ or 10.065.

EXAMPLE 2:

Suppose we collect y height and x highest level of education on 2000 subjects randomly selected in Stockholm and ran a linear regression for the association between y and x. We first have to convert education level to a set of dummy variables $\{x_{ba}, x_{ma}, x_{phd}\}$ where each are binary variables that take on 1 if this is the highest level of education and 0 otherwise. Everyone we spoke to had at least a BA. Below are the results from running the following model.

$$\text{E(height}|x_{ma}, x_{phd}) = \beta_0 + \beta_1 * x_{ma} + \beta_2 * x_{phd}$$

	Estimate	Std. Error	t value	P-value
$\widehat{\beta}_0$	166	10.45	15.89	<0.001
$\widehat{\beta}_1$	5	3.25	1.538	0.124
$\widehat{\beta}_2$	22	5.4	0.158	<0.001

Table 7: The example coefficient matrix from a linear regression with a single categorical predictor.

This means that in subjects that have a BA as their highest degree, the mean of y is β_0 and in subjects that have a MA as their highest degree, the mean of y is $\beta_0 + \beta_1$ and and in subjects that have a MA as their highest degree, the mean of y is $\beta_0 + \beta_2$. This means that β_1 is the difference in expected height in two groups that differ in what their highest degrees is from BA to MA, and β_1 is the difference in expected height in two groups that differ in what their highest degrees is from BA to PhD. [70]

[70] just because the categorical variable has a natural ordering, the mean for PhD subjects is NOT $\beta_0 + \beta_1 + \beta_2$

The p-values are for the null hypotheses

H0: $\beta_0 = 0$

H0: $\beta_1 = 0$

H0: $\beta_2 = 0$

Because the p-value associated with β_0 is <0.001, we reject the null hypothesis at the 0.05 level that $\beta_0 = 0$, but what does this mean really? We have evidence to support the hypothesis that the mean height is not zero for people with a BA...Great! that is not really interesting. The p-value associated with β_1 is 0.124, which means we cannot reject the null $\beta_1 = 0$, meaning that we do not have evidence to support that the DGM expectation of heights in people with BAs is different than those with MAs. Again, we cannot say that this is evidence of the null. However, there is evidence to reject the null that $\beta_2 = 0$ and that the DGM expectation of heights for those people with

a PhD differs from that of people with BAs.

Based on our estimates 166 cm is the expected height of people with BAs, 166+5=171 cm is the expected height of people with MAs and 166+22=188 cm is the expected height of people with PhDs. $\widehat{\beta}_{x_{ma}}$, 5, is the estimated difference in the expected values between groups that differ by their highest education level BA to MA and $\widehat{\beta}_{x_{phd}}$, 22, is the estimated difference in the expected values between groups that differ by their highest education level BA to PhD.

Yes it is a little strange to think of x causing y in this case, and if you were to draw a DAG[71], there could be an arrow from y here to x, but clearly not the other way. This however is a fine model for Step 1, when you might just be looking to generate some ideas, or if your true hypothesis is association. So we can simply state that we do or do not have evidence to reject an association between x and y. [72]

[71] Can you think of a reason that that education level might be associated with height, but height simply being a mediator or height and education level being confouded by something?

[72] As there are multiple levels of education level here, we may wish to do a test that education level was associated in general with y, in this case this is the same as the overall p-value, via F test for the regression, as only 1 categorical variable was included. But in later sections we will add more variables and this may not be the case. We will cover both of these tests in lab

Example 3: We observe the SBP for 1000 subjects, y, and their age in years, x, and run a linear regression to look for an association between them. Below are the results from running the following model.

$$\text{E(sbp|age)} = \beta_0 + \beta_1 * \text{age}$$

	Estimate	Std. Error	t value	P-value
$\widehat{\beta}_0$	105	10.98	9.56	<0.001
$\widehat{\beta}_1$	0.5	.13	3.85	<0.001

Table 8: The example coefficient matrix from a linear regression for a single continuous predictor.

This means that, when age is 0, the expected values of the DGM of SBP for age 0 people is β_0. This isn't a useful value alone anymore, but we need it to make β_1 make sense. The expectation of DGM for SBP for people of age x is $\beta_0 + \beta_1 * x$. This also means that β_1 is the difference in expected values of the DGM of SBP for any two groups that differ in age by one year.

The p-values are for the null hypothesis

H0: $\beta_0 = 0$

H0: $\beta_1 = 0$

For both, we have evidence to reject the null at the 0.05 level. Meaning that we have evidence to reject that the Expected value of the DGM of SBP for people of age 0 is 0, and that we have evidence to reject that SBP is not associated with age. This p-value will be identical a Pearson's linear correlation test. We estimate that the expected difference in the expected value of the DGM of SBP for two groups that differ in age by 1 year is 0.5 millimeters of mercury. This means that regardless of the years of age increasing by 1 year is expected to

be associated with a 0.5 unit increase is SBP. This is why this type of regression is called linear! [73].

[73] As we will see below, this is an old term and we can in fact transform the data to estimate non-linear relationships

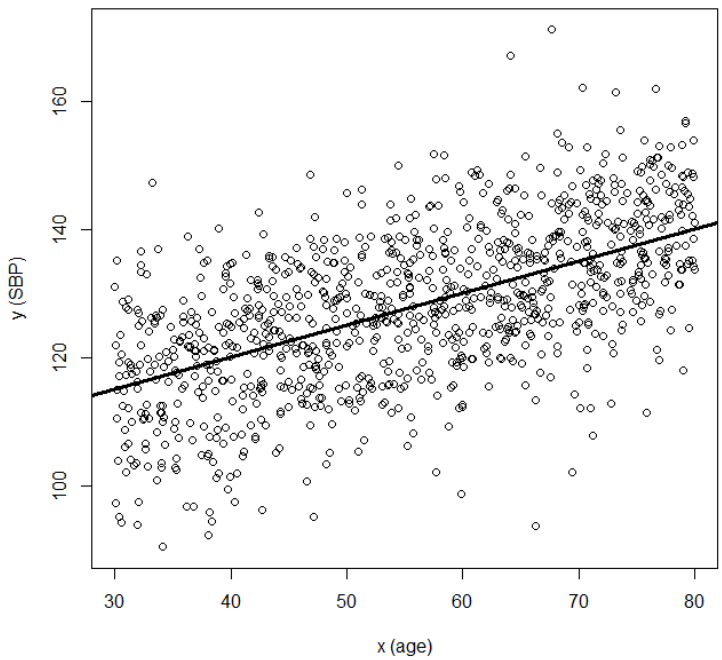


Figure 26: Linear regression of SBP on Age, assumes that for a one unit change in age, there is always the same expected difference in SBP.

Consider what this means. comparing a group of 100 years old to a group of 101 years old, we expect to see a 0.5 difference in SBP, and that same difference is expected from 50 to 51 or 10 to 11...and based on our assumptions for this model we believe this is the true DGM of SBP.

Predictions from these models can simply be obtained using the estimated values. For example the predicted expected SBP value for a group of age 20 is $105 + 0.5 * 20$ or 115. For a group age 35 it is $105 + 0.5 * 35$ or 122.5 and so on. In the first Example the predicted expected Temperature for student 1 is 10.056 and for student 2, $10.056 + 0.009$, 10.065. As we add more variables to the models this methods of obtaining predictions for the particular levels of the covariates will not change.

Robust Standard Errors

ROBUST MEANS DIFFERENT THINGS IN DIFFERENT CONTEXTS. Sometimes is means robust to outliers, sometimes model missspecification and sometimes, like here, it means allowing for $\text{Var}\{\epsilon|x\} = \sigma^2(x)$ to depend on x, or robust to violations of homoscedasticity. Using this robust estimate of the standard errors reduces the assumptions needed to simple linear regression by 1, but also changes one of these assumptions.

ASSUMPTIONS FOR CORRECT INFERENCE:

1. The pairs (y_i, x_i) are iid
2. Exogenous error $E\{\epsilon|x\} = 0$
3. Large enough sample
4. Correct specification. The linear functional form must coincide with the form of the actual data-generating process.

UNDER THESE ASSUMPTIONS one can use one of several proposed heteroscedasticity-consistent estimators of the variance of $\widehat{\beta}$. One example of this is White's (1980), is often referred to as White's and is given by[74]:

$$\widehat{\text{Var}}(\underline{\widehat{\beta}}) = (\underline{x}^T * \underline{x})^{-1}(\underline{x}^T \text{diag}(\epsilon_i^2, \ldots \epsilon_n^2)\underline{x})(\underline{x}^T * \underline{x})^{-1}$$

Where $\text{diag}(\epsilon_i^2, \ldots \epsilon_n^2)$ is an n-by-n matrix with ϵ_i^2 on the diagonal and 0s everywhere else. For example if n=2 then

$$\text{diag}(\epsilon_i^2, \ldots \epsilon_n^2) = \begin{bmatrix} \epsilon_1^2 & 0 \\ 0 & \epsilon_2^2 \end{bmatrix}$$

MORE IMPORTANTLY almost all statistical programs now easily implement some type of heteroscedasticity robust standard error estimates. When you have an estimator that is almost no loss, because you were assuming a large sample size anyway, why not use robust standard errors. Like in the t-test why would you ever assume that you had equal variance, when you could avoid this assumption. Here you must assume a large sample size, so in smaller samples, there is a reason to use model based standard errors. In addition, although you are avoiding inconsistent standard error estimates by using robust SE, they can lead to lower power, which maybe why people continue to not use them even in large enough sample sizes.[75]

[74] you don't need to know this! but note the shape, something in the middle with two things on the outside that are the same. The outside bit is often called bread, and over many different versions of these changes very little, what changes more often is the middle and that is called meat, or cheese or tofu by some vegetarian statisticians. This is why you have probably heard the term 'sandwich' variance estimator.

[75] This is not a very good reason not to do it! In fact, sometimes they may lead to higher power!

Multivariate linear regression

As noted in the previous pages, a linear regression with a single predictor x will have identical inference to:

- A t-test with equal variance, if x is binary
- ANOVA, if x is categorical (which is why you never ever need to use ANOVA)
- Pearson's linear correlation test, if x is numerical

It seems that linear regression is just doing the same things as other tests. However, in regression (not just linear) we can do a wonderful thing: add more variables at once! When you add more than one variable to a model you must make one more assumption. The assumption when using model based standard errors are now:

Assumptions for correct inference and interpretation:

1. The pairs (y_i, x_i) are iid
2. Exogenous error $E\{\epsilon|x\} = 0$
3. $\text{Var}\{\epsilon|x\} = \sigma^2$, homoscedasticity, note that this does not depend on x
4. Large enough sample **or** $\epsilon|x \sim \text{N}(0, \sigma^2\mathbb{I})$
5. Lack of perfect multicollinearity in the predictors or independent variables, meaning there is not a perfect correlation between any two covariates and there is enough data to distinguish between the levels of the included variables. This is sometimes called the full rank assumption, which implies that $(\underline{x}^T\underline{x})^{-1}$ has a finite value, which is always true in the case of 1 predictor, but may not be true if two predictors are perfectly correlated.
6. Correct specification. The linear functional form must coincide with the form of the actual data-generating process.

Example 1: Suppose you are attempting to generate hypotheses by looking for associations between blood pressue and a set of covariates. Your response variable (y) is systolic blood pressure (SBP), and assume you have 3 predictors: sex (M/F), smoking (never/former/current), and age in years. Consider the following models:

- Model 1: E(SBP|sex) = β_0 + β_1*sexM

- Model 2: E(SBP|Smoking) = β_0 + β_1*smokeF + β_2*smokeN
- Model 3: E(SBP|age) = β_0 + β_1*age
- Model 4: E(SBP|Smoking, age, sex) = β_0 + β_1*sexM + β_2*smokeF + β_3*smokeN + β_4*age

Here, sexM is 1 for males, and 0 for females; smokeF is 1 for former smokers, and 0 otherwise; and smokeN is 1 for never smokers, and 0 otherwise. Models 1,2, and 3 are interpreted as described in the previous pages. In model 4, you have all covariates at once.
Suppose starting with model 1 you are investigating the mean SBP in Males and Females, and you are interested in the coefficient of the sexM variable. What is the difference between the value of β_1 in Model 1 and in Model 4?
The value of β_1 in Model 1 and that in Model 4 will generally be different.

- In Model 1, β_1 is the difference between the expected value of SBP of males, and the expected value of SBP of females.
- In Model 4, β_1 is the difference between the expected value of SBP of a group of males and that of a groups of females having the same age and smoking habits.
- In Model 1, we say that β_1 is the 'unadjusted' or 'crude' gender difference.
- In Model 4, we say that β_1 is the 'adjusted' gender difference. To be precise, it is adjusted for age and smoking habits.

β_0 differs between the two models as well. The value of β_0 in Model 1 and that in Model 4 will generally be different.

- In Model 1, β_0 is the expected value of SBP for females
- In Model 4, β_0 is the expected value of SBP for females, that currently smoke and that are of age 0.

Consider model 2. You are investigating the association between SBP and smoking habits, and you are interested in the coefficient of the smokeF, and smokeN variables. What is the difference between Model 2 and in Model 4?
Starting with the intercept this time, β_0 differs between the two models. The value of β_0 in Model 2 and that in Model 4 will generally be different.

- In Model 2, β_0 is the expected value of SBP for current smokers

- In Model 4, β_0 is the expected value of SBP for females, that currently smoke and that are of age 0.

The coefficient for smokeF in Model 2 and that in Model 4 will generally be different.

- In Model 2, coefficient for smokeF is the difference between the expected value of SBP of former smokers, and the expected value of SBP of current smokers.
- In Model 4, coefficient for smokeF is the difference between the expected value of SBP of former smokers, and the expected value of SBP of current smokers of the same age and sex.
- In Model 2, we say that coefficient for smokeF is the 'unadjusted' or 'crude' smoking difference.
- In Model 4, we say that coefficient for smokeF is the 'adjusted' smoking difference. To be precise, it is adjusted for age and sex.

Similarly, The coefficient for smokeN differs in interpretation. In Model 2, coefficient for smokeN is the difference between the expected value of SBP of never smokers, and the expected value of SBP of current smokers, and in Model 4, coefficient for smokeN is the difference between the expected value of SBP of never smokers, and the expected value of SBP of current smokers of the same age and sex. Model 4 is adjusted for sex and age and model 2 is not.
Consider model 3. You are investigating the association between SBP and age, and you are interested in the coefficient of age. What is the difference between Model 3 and in Model 4?
Starting with the intercept, β_0 differs between the two models. The value of β_0 in Model 3 and that in Model 4 will generally be different.

- In Model 2, β_0 is the expected value of SBP for people of age 0.
- In Model 4, β_0 is the expected value of SBP for females, that currently smoke and that are of age 0.

The coefficient for age in Model 3 and that in Model 4 will generally be different.

- In Model 3, coefficient for age is the difference between the expected value of SBP for groups that differ in age by 1 year.
- In Model 4, coefficient for age is the difference between the expected value of SBP for groups that differ in age by 1 year but have the same smoking habits and sex.

- In Model 3, we say that coefficient for age is the 'unadjusted' or 'crude' age difference.
- In Model 4, we say that coefficient for age is the 'adjusted' age difference. To be precise, it is adjusted for smoking habits and sex.

IN EACH CASE the adjusted coefficient can be said to the difference or expected change for a 1 unit change in the variable it is associated with, **holding the other variables constant**.

In model 4:

- β_0 is the expected value of SBP for females, that currently smoke and that are of age 0.
- β_1 is the change in the expected value of SBP between a group of men and a group of women, holding age and smoking habits constant.
- β_2 is the change in the expected value of SBP between a group of former smokers and a group of current smokers, holding age and sex constant.
- β_3 is the change in the expected value of SBP between a group of never smokers and a group of current smokers, holding age and sex constant.
- β_4 is the change in the expected value of SBP for a 1 unit change in age, holding smoking habits and sex constant.

CONSIDER THIS SITUATION: Males smoke more than females (i.e., there are more current and former smokers among males). Also, suppose smoking causes the mean SBP to increase and that men have higher SBP naturally as well. The DAG for this is as follows.

$$\begin{tikzcd}
 & \text{Sex} \arrow[dl] \arrow[d] \\
\text{smoking} \arrow[r] & \text{SBP}
\end{tikzcd}$$

YOU ARE INTERESTED IN THE EFFECT ON SEX ON SBP. In Model 1, β_1 represents the unadjusted difference between the mean SBP of males and that of females. Suppose you find $\widehat{\beta}_1 = 12$. The observed gender difference is the result of two effects:

- Males are 'biologically' different from females, and tend to have a higher SBP

- Males smoke more than females, which generates an additional gap in terms of mean SBP.

In this case, smoking is a mediator of the sex effect on SBP. If you care about the full or total sex effect, model 1 is the correct model to fit. If you want to know the direct effect of sex on SBP that is not mediated by smoking Model 4 is the correct model to fit.

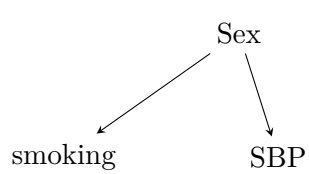
YOU ARE INTERESTED IN THE EFFECT ON NEVER SMOKING AS COMPARED TO CURRENT SMOKERS ON SBP. In Model 2, β_2 represents the unadjusted difference between the mean SBP never smokers and that of current smokers. Suppose you find $\widehat{\beta}_2 = 10$. The observed gender difference is the result of two effects:

- the impact of smoking on SBP
- Males smoke more than females and males have higher blood pressure.

in this case sex is a confounder of the smoking SBP relationship, and unless you are only interested in looking for an association of any type, sex must to included in the model for proper inference and estimating of the effect of smoking on SBP. Model 4 is always the correct model in this case.

NOTE that the correct model depends on the covariate of interest, often referred to in epidemiology as the exposure. For the effect of sex on SBP model 1 or 4 maybe correct, while if you are interested in the effect of smoking, under this DAG, only model 4 is correct. It should be clear that in order to determine what model to use, one must have a DAG, at least in mind as well as an effect of interest.

CONSIDER THIS SITUATION: Males smoke more than females, but smoking has no effect on SBP and males have higher SBP.



YOU ARE INTERESTED IN THE EFFECT ON NEVER SMOKING AS COMPARED TO CURRENT SMOKERS ON SBP In Model 2, β_2 represents the unadjusted difference between the mean SBP never smokers and that of current smokers. Suppose you find $\widehat{\beta}_2 = 10$. The

observed smoking difference is the result of the confounding by the sex effect.
Unlike the setting above where there is an effect of smoking and sex making the effect hard to estimate. Here there is no effect and yet, you will often reject the null of no association. This is the scenario that is most troubling in observational studies!

CONSIDER THIS SITUATION: the mean SBP increases with age for males, but decreases with age for females. The DAG for this is as follows.

$$\begin{tikzcd} & \text{age} \arrow[d] \\ \text{Sex} \arrow[r] & \text{SBP} \end{tikzcd}$$

SUPPOSE WE ARE INTERESTED IN THE IMPACT OF AGE ON SBP
In Model 3, β_1 represents the unadjusted difference between the mean SBP for a 1 unit change in change ignoring sex and smoking. Suppose you find $\widehat{\beta}_1 = 0$. The observed gender difference is the result of two effects:

- the impact of age on SBP in males
- the impact of age on SBP in females

And in this case they cancel each other out. This is still a fine model as this is the marginally effect on age on SBP in the population, as males and females are roughing 50% each. However, if you wish to know if age has an effect on SBP within subgroups of sex, we need to fit a different model. Model 4 is not the correct model here, as holding sex constant, we do not have one constant effect of age. The correct model is instead.

- Model 5: E(SBP|Smoking, age, sex) = β_0 + β_1*sexM + β_2*age + β_3*sexM*age

THIS MODEL HAS A SLIGHTLY DIFFERENT INTERPRETATION than an adjusted model. We are no longer simply holding the other variables constant, as that would be impossible without holding the other two variables constant in the case of sexM*age, and if age and sexM is constant so it sexM*age. For this reason. Figure 27 depicts a much more reasonable scenario where age impacts SBP, but simply has less of an effect in females.

In model 5:

- β_0 is the expected value of SBP for females of age 0.
- β_1 is the change in the expected value of SBP between a group of men and a group of women, who are age 0.
- β_2 is the change in the expected value of SBP for a 1 unit change in age, for females i.e. sexM=0
- β_3 is additional change in the expected value of SBP for a 1 unit change in age, for males versus females. OR is the additional change in the expected value of SBP between a group of men and a group of women, for a 1 unit change is age.

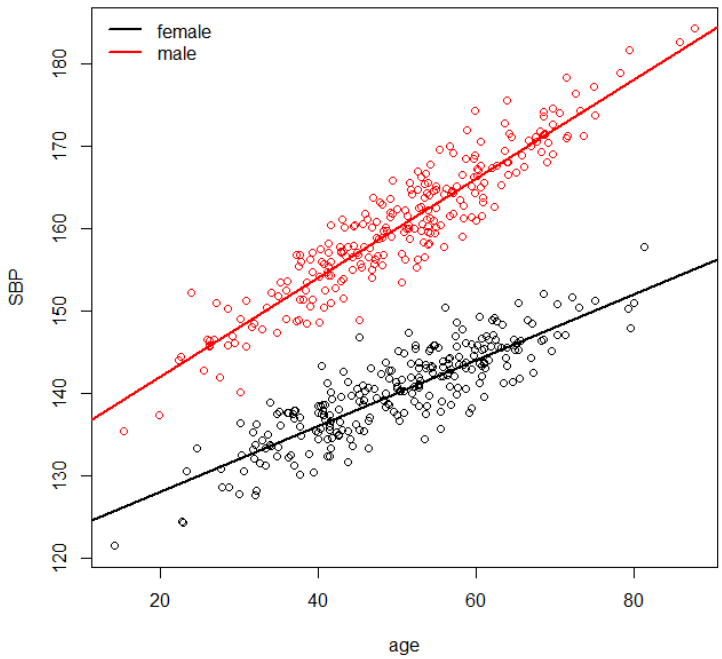


Figure 27: For an interaction model the slope over age can be different between males and females. Here they don't go in opposite directions, but the females slope is much lower.

In this case the slope of age is not the same in males and females, but also the effect of sex differs by age. We probably should not describe the effect of age by a single number, because the effect of age is different in males and females. And we probably should not describe the gender difference by a single number, because the gender difference changes with age. It goes both ways! So when you describe it, as in

the table above, focus on the effect of interest. If, as in this example you were interested in the effect of age, describe it that way, as the additional effect on age, in males versus females.

IF THERE IS NOT AN INTERACTION EFFECT, β_2 IN THIS MODEL WILL BE NEAR ZERO, AND THE MODEL WILL COLLAPSE BACK TO:

- Model 6: E(SBP| age, sex) = β_0 + β_1*sexM + β_2*age

In this model there are different intercepts for males and females, suggesting they differ at birth, but the effect of age on SBP is the same in both groups; Figure 28 illustrates this result.

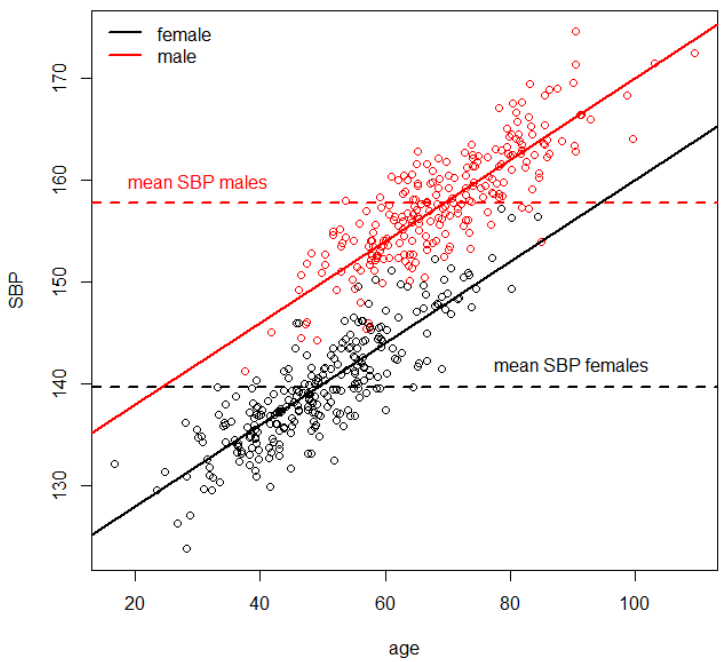


Figure 28: For an adjusted model gives the same slope over age regardless of sex, but allows for different intercepts by gender.

Transformations

LINEARITY IS PRETTY RESTRICTIVE so one can transform covariates to relax this assumption. For example if you think age might have a U-shape effect on SBP meaning from age 0 to 50 it is increasing the 50 to 100 it is decreasing, you can subtract 50 from each observed age, and include this new variables, and the square of it the model.
You can describe a nonlinear effect of age and in particular a u-shape by using the model: E(SBP|age) = β_0 + β_1*(age-50) + β_2*(age-50)^2
This is still a linear regression! However, it describes a non-linear function of age, and more precisely a parabola or U, with the middle at 50. If you don't subtract 50 the middle of the u will be at age 0, which doesn't make it a U over the possible range of ages.
If a parabola is not flexible enough, you may use a cubic function with again centering at a value like 50:

E(SBP|age)= β_0 + β_1*age + β_2*age^2 + β_3*age^3

this will give you a sideways s-shape curve.

you can add any other power of age, e.g., age4, age5 or the square root of age, its logarithm, or any other transformation. [76] Because the logarithm transformation is used very often in Epidemiology below we go over this transformation of a covariate or the outcome in detail.

[76] A method to define flexible models is to use "splines", which are very flexible polynomial functions.

Log Transformations of Predictors

TO REVIEW LOGS. The logarithm function, often abbreviated log, is the inverse function to exponentiation. If there is no subscript in statistics on the log function, this stands for the natural log $\log_e$, where e is as before Euler's number. If there is a subscript then $\log_k$ is log base k and means $\log_k(x) = y$ if $k^y = x$

You don't need to worry so much about the base, using different bases for logarithms is just like measuring length in different units (cm, meters, KM, light years). No matter what base you use $\log(1) = 0$ because $k^0 = 1$ for all k and $\log(0)$ is $-\infty$, and log of anything less then 0 is undefined.

Based on this, note that a 1 unit difference in $\log_k(x)$ is equal to a k-fold increase in x, and a c unit difference in $\log_k(x)$ corresponds to a k^c-fold increase in X. log transformations of medical data are common.

ALSO NOTE THAT WHAT WE CALL THE SAMPLE MEAN ABOVE IS ALSO CALLED THE ARITHMETIC MEAN, THERE ARE OTHER MEANS! In particular, the geometric mean is often used for lab values like viral load. The geometric mean of a sample of size n from x is:

$$\text{geomean}(x) = e^{\left[\frac{1}{n}\sum log_e(x)\right]}$$

This e raised to the arithmetic mean or sample mean of the log transformed xs. Because logs spread out the lower end of the curve near 0 and bunch together the higher end of the curve large numbers, the geometric mean tends to be less sensitive to outliers.

Suppose we model

$$\text{E}[Y|X = x] = \beta_0 + \beta_1 \log_k(x)$$

- From our standard interpretation of regression slope parameters, we know that every 1 unit difference in $\log_k(x)$ is associated with a β_1 unit difference in the expected value of Y. And that every c unit difference in $\log_k(x)$ is associated with a $c\beta_1$ unit difference in the expected value of Y.

- If we want to talk about a 10% increase in x, then that would correspond to a c = $\log_k(1.1)$ unit increase in $\log_k$(x).

- Suppose we model predictor VL on a log base 10 scale. Because we never see a 10-fold increase in VL, when interpreting our model parameters it might be better to consider comparisons between populations which differ in VL by, say, 10%. We would then estimate the difference in the expected response as $\log_1 0(1.1)\widehat{\beta}_1$, where $\widehat{\beta}_1$ was the least squares estimate for the slope parameter in the regression. Note that we would find a confidence interval for the effect associated with that 10% change in VL by multiplying the CI for β_1 by $\log_1 0(1.1)$ as well. Then a 1 unit change in your predictor corresponds to a 10% change in VL.

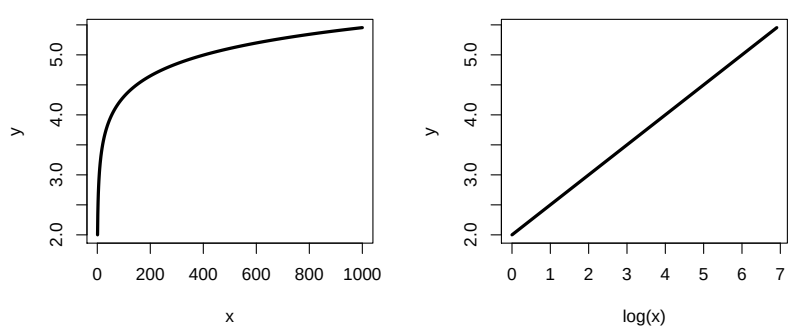


Figure 29: The estimated expected value of y will be linear on the log scale, but will not be linear on the original x scale. x has a range 0 to 1000, while the range of $\log_e$(x) is much smaller.

Transformation of Response

Suppose we model (for arbitrary base j)

$$\mathrm{E}[\log_j(y)|x] = \beta_0 + \beta_1 x$$

- Using the standard interpretation of regression slope parameters, we know that every 1 unit difference in x is associated with a β_1 unit difference in the expected value of $\log_j$(y), and every c unit difference in x is associated with a $c * \beta_1$ unit difference in the expected value of $\log_j$(y).
- Unfortunately, a β_1 unit difference in the expected value of $\log_j$(y) does not have an easy interpretation in the expected value of y. However, statements made about the distribution of $\log_j$(y) are generally not well understood by the general population, so we need to find another way.
- The expected value of $\log_j$(y) is the $\log_j$ of the geometric mean of Y. Thus, we can make statements about the geometric mean of Y considering our model to be

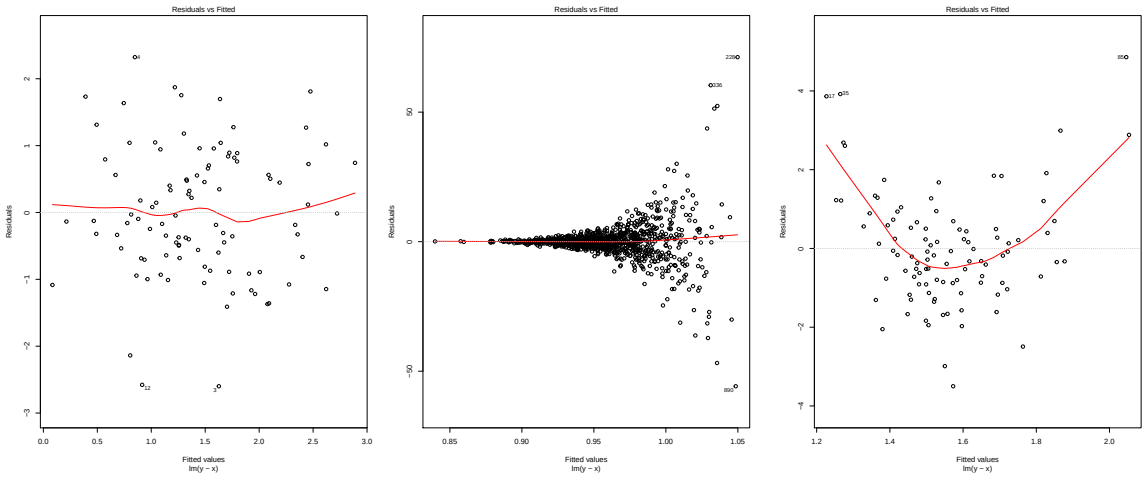
$$\mathrm{E}[\log_j(y)|x] = \log_j(\mathrm{Geomean}[y|x]) = \beta_0 + \beta_1 * x$$

- Under this modification, a β_1 unit difference in the base j logarithm of the geometric mean of y corresponds to a j^{β_1}-fold change in the geometric mean of y. Similarly, a $c * \beta_1$ unit difference in the base j logarithm of the geometric mean of y corresponds to a $\mathrm{j}^{c\beta_1}$-fold change in the geometric mean of Y . We can say that $\mathrm{j}^{c\beta_1}$ is the ratio of geometric means for two populations which differ by c units in their values for x.

Model checking

Assuming you are in step 1 you should check the models you are using to make sure the hypothesis you are generating are not driven by a poor model fit. If you are in step 3, you will have already specified the model to use, and although you can check it, you cannot really change your mind about what to use once it is has been run. However, model checking does allow you to be sure your conclusion, particular those of rejecting the null are not being driven by violation of the assumptions. One way to do this is to look at a plot of the residuals versus the fitted values from the model. This plot should look random and equally distributed over the fitted values. If you were to use a smoother to fit a line to these data, like a lowess curve, you should see something approximately flat about zero. Divisions to this

can mean evidence of violation of the assumptions, as is demonstrated in the plots below.



: A : B : C

Figure 31: Residual versus Fitted value plots to check assumptions

Panel A of Figure 31 illustrates what a residual versus fitted value plot would Ideally look like. There should be an almost straight line at 0, with points equally spaced away from the line across the full range of the fitted values. Panel B illustrates what a residual versus fitted value plot looks like when there is strong heteroskedasticity, i.e. the variance of residuals depends on x. Panel B illustrates what a residual versus fitted value plot looks like when there is a strong non-linear relationship between x and y, but the model was fit assuming a linear relationship. There are of course other things that would be concerning shapes in a residual versus fitted value plot, but the above are clear violations. One should also not be too sensitive, as all models are wrong. Provided that there is no clear evidence of a violation, very clearly not straight lines or a lowess not centered zero,

THE OTHER PLOT to look at is a QQ-plot which looks at the approximate normality. However, this is only really need in smallish samples. There are also tests one can use to detect deviations from normality, but as mentioned above a Shapiro-Wilk test has a null of normality, like almost all tests one could use testing normality, and will have low power in small samples. As small samples are the only place where normality actually matters, this seems close to pointless. In large samples tests for normality will almost always reject, but because you have a large sample you don't need to worry about it, provided you don't see evidence of other assumption violations.

Comparing models and covariate selection

IF YOU ARE IN STEP 1 then comparing many different models and/or doing univariate regression for all possible covariates and including those that are significant in a multivariate model with the covariate of interest is just fine. In step 1 any form of machine or human learning for covariate selection or model fit is perfectly fine.

IF WE SIMPLY WANT TO COMPARE TWO LINEAR MODELS and if they are nested we can use a likelihood ratio test after running the two models. We will not going into the details of this, but most software packages have a function for doing this test and we will cover it in lab. An example of a nested models and non-nested models are given here:

Nested
Model 1: includes age, sex, BMI
Model 2: includes age, sex, BMI, smoking

Not Nested
Model 3: includes age, sex, BMI
Model 4: includes age, sex, smoking

Model 2 contains Model 1, plus an additional variables smoking.
Model 3 and Model 4 simply have different sets of variables.

IF THE MODELS ARE NOT NESTED there are three common measure of model fit that can be compared. AIC, BIC and R^2 (or adjusted R^2). R^2 and adjusted R^2, only really make sense for linear regression, and at least R^2 should be very familiar in simple linear regression because, $\sqrt{R^2} = \rho$ The Pearson's correlation coefficient. R^2 can be problematic because adding more variables will never decrease, and will often increase, R^2. Therefore you can get a higher R^2 by just adding all covariates. Instead, adjusted R^2 attempts to account for an increased number of covariates. For this reason adjusted R^2 is always preferred if the models do not contain the same number of variables, with large adjusted R^2 or R^2 being better. AIC and BIC are similar goodness of fit measures that indicate the better model when they are lower. We will again to detail how these are calculated, but they are commonly used and are available in most software packages. They both can be used to compare not nested models. Both penalize for increasing numbers of covariates and in general increasing complexity, in particular BIC tends to penalize more heavily for complex models. There is no clear evidence that AIC is better than BIC, and when

they disagree, there is no clear reason to select one over the other. You may wish to use all three in combination.

When comparing two not nested models one of which has the lower AIC and higher adjusted R^2, and the other that have the lower BIC, you might want to go with the 1st. Other criteria could be introduced as well, such as you always prefer a less complex model, unless all three measures are better for the more complex, for example. Comparison of non-nested models is more of an art than a science, and it also matters what you wish to use the model for.

To choose between two regression models:

- If the models are NOT nested, you can use R^2 or adjusted R^2 the better model has higher value, or AIC or BIC for which the better model has smaller value.
- If the models are nested, simply use likelihood ratio test (available for most regression methods). Which actually gives you a test to compare two models.

If we are in step 3, you should NEVER do this. If you are in step 3 and you believe that you may have some form of confounding, you should draw a DAG to illustrate your belief of the relationship and pre-specify the covariates that will be included in the model and how, PRIOR to looking at any of the data. If there is no way to do this, or if changes need to be made prior to running, but after looking at the data, look at the covariates alone, without looking at the outcome or the exposure, and make your decisions. Once you have looked at the y, x, and covariate relationship, and used that in your decision making process, you no longer have type one error rate control and that is almost always what we mean by 'valid' inference. So you cannot fit two different models, nested or not, to draw conclusions about the x, y relationship once we are in step 3, because in fitting two and picking one, you will not control the type I error rate, and inference based on the standard errors that do not account for this extra step is questionable.[77]

[77] In reality, if it was just two models, and it was selected by BIC, for example, it might not be that bad. If it was selected by the lower p-value for the coefficient of x on the other hand, doing this at all will cause unbound type I error rate inflation. This means in large enough samples sizes the type I error will approach one, just as power does.

Index

chi-square test, 70

Distribution Parameters, 43

license, 4

odds ratio, 69

permutation test, 71

risk difference, 69

risk ratio, 69

Two sample tests, 73